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# Tumour vasculature at single-cell resolution

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Tumours can obtain nutrients and oxygen required to progress and metastasize through the blood supply[1]. Inducing angiogenesis involves the sprouting of established vessel beds and their maturation into an organized network[2,3]. Here we generate a comprehensive atlas of tumour vasculature at single-cell resolution, encompassing approximately 200,000 cells from 372 donors representing 31 cancer types. Trajectory inference suggested that tumour angiogenesis was initiated from venous endothelial cells and extended towards arterial endothelial cells. As neovascularization elongates (through angiogenic stages SI, SII and SIII), *APLN*$^{+}$ tip cells at the SI stage (*APLN*$^{+}$ Tip$^{SI}$) advanced to Tip$^{SIII}$ cells with increased Notch signalling. Meanwhile, stalk cells, following tip cells, transitioned from high chemokine expression to elevated *TEK* (also known as *Tie2*) expression. Moreover, *APLN*$^{+}$ Tip$^{SI}$ cells not only were associated with disease progression and poor prognosis but also hold promise for predicting response to anti-VEGF therapy. Lymphatic endothelial cells demonstrated two distinct differentiation lineages: one responsible for lymphangiogenesis and the other involved in antigen presentation. In pericytes, endoplasmic reticulum stress was associated with the proangiogenic *BASP1*$^{+}$ matrix-producing pericytes. Furthermore, intercellular communication analysis showed that neovascular endothelial cells could shape an immunosuppressive microenvironment conducive to angiogenesis. This study depicts the complexity of tumour vasculature and has potential clinical significance for anti-angiogenic therapy.

Tumour angiogenesis is a critical hallmark of cancer. It is induced to help tumours obtain the required nutrients and oxygen to sustain growth[1]. Angiogenesis can occur during any stage of tumour progression[4] and is manifested by the formation of new vessels from the existing vascular network[2,3]. Tumour vasculature, compared with normal blood vessels, shows increased permeability, irregular morphology and poor organization[2]. Endothelial cells (ECs) and mural cells (MCs) are primary vasculature components that are directly involved in tumour angiogenesis[5]. EC subpopulations are quite heterogeneous, encompassing distinct vascular-type and organ-specific characteristics[6,7]. The plasticity of ECs makes them a target for anti-angiogenic therapies (AATs), dedicated to achieving tumour vascular normalization by blocking angiogenic signalling[8]. Combining AAT with tumour immunotherapy has been shown to generate more potent anti-tumour effects in preclinical studies[9,10]. Taking advantage of single-cell profiling technologies, we generated a pan-tumour vascular landscape covering around 200,000 vascular cells and unravelled the intricacies of tumour vasculature.

## Single-cell atlas of tumour vasculature

To dissect the tumour vascular microenvironment (TVM), we performed in silico sorting of ECs and MCs in both in-house and public datasets of 437 tumour samples and adjacent non-tumour (ANT) samples from 372 donors (Fig. 1a,b and Supplementary Table 1). A total of 183,977 TVM cells were subjected to rigorous quality control, and 24.2% of which were newly sequenced in this study (Fig. 1c). We integrated data of these cells, generated from multiple platforms, and corrected for batch effects (Methods and Extended Data Fig. 1). Subsequently, we annotated the cell types on the basis of representative gene signatures (Fig. 1c,d and Supplementary Table 2). Within tumour tissues, ECs were predominantly classified as lymphatic ECs (LECs) (characterized by *LYVE1* and *PROX1*) and vascular ECs (VECs), including arterial ECs (ArtECs) (characterized by *SEMA3G*), venous ECs (VenECs) (characterized by *ACKR1*) and capillary-like ECs (CapECs) (characterized by *RGCC*). Moreover, we detected VECs undergoing endothelial-to-mesenchymal

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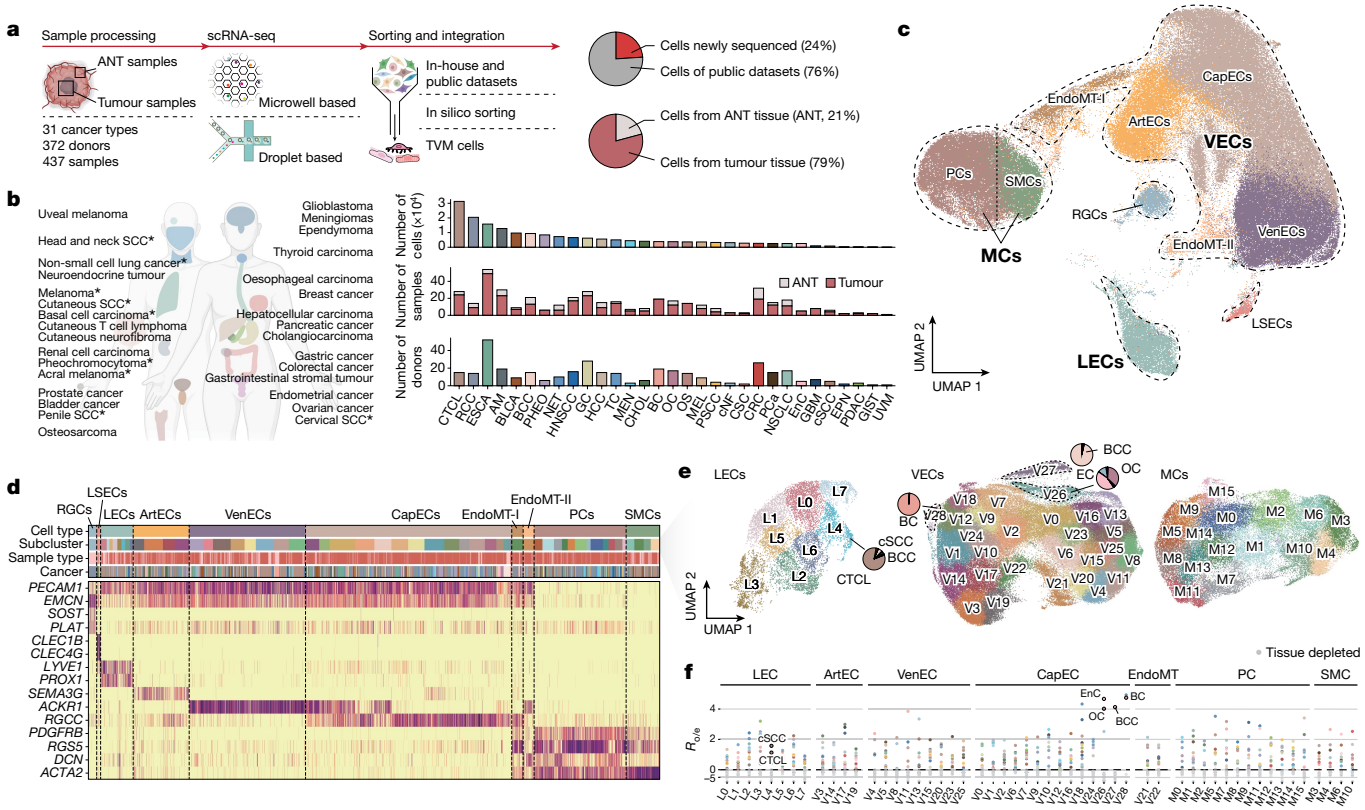


**Fig. 1 | Cell populations of the tumour vasculature. a**, Left, schematic of the data generation, study design and integration of the atlas. Right, the proportion of TVM cells from internal/public datasets (top) and ANT/tumour tissues (bottom). **b**, Cancer types covered in the present study. The asterisks indicate cancer types with newly generated data. The bar plot shows the number of detected cells (top), samples (middle) and donors (bottom) for each cancer type. **c**, UMAP analysis of all TVM cells, coloured by cell types. **d**, The expression levels of representative signature genes. **e**, UMAP visualization of TVM subclusters. The pie charts show the proportion of cancer types in subclusters of similar tissue origin. **f**, The $R_{o/e}$ value (log scaled) of cancer types in each TVM subcluster. Subclusters of similar tissue origin are labelled by cancer type. Gynaecologic: endometrial cancer and ovarian cancer; mammary: BC; skin: basal cell carcinoma, cutaneous T cell lymphoma and cutaneous squamous cell carcinoma. AM, acral melanoma; BC, breast cancer; BCC, basal cell carcinoma; BLCA, bladder cancer; CHOL, cholangiocarcinoma; cNF, cutaneous neurofibroma; CRC, colorectal cancer; CSC, cervical squamous cell carcinoma; cSCC, cutaneous squamous cell carcinoma; CTCL, cutaneous T cell lymphoma; EnC, endometrial cancer; EPN, ependymoma; ESCA, oesophageal carcinoma; GBM, glioblastoma; GIST, gastrointestinal stromal tumour; GC, gastric cancer; HCC, hepatocellular carcinoma; HNSCC, non-small cell lung cancer; MEL, melanoma; MEN, meningioma; NET, neuroendocrine tumour; NSCLC, non-small-cell lung cancer; OC, ovarian cancer; OS, osteosarcoma; PCa, prostate cancer; PDAC, pancreatic cancer; PHEO, pheochromocytoma; PSCC, penile squamous cell carcinoma; RCC, renal cell carcinoma; TC, thyroid carcinoma; UVM, uveal melanoma. The diagrams in **a** and **b** were created using BioRender.

transition (EndoMT), which co-expressed EC and mesenchymal markers, and further distinguished these cells as *PDGFRB*$^{+}$ EndoMT-I and *DCN*$^{+}$ EndoMT-II cells (Extended Data Fig. 2a). Two types of MCs were identified, including pericytes (PCs) (characterized by *RGS5* and *PDGFRB*) and smooth muscle cells (SMCs) (characterized by *ACTA2* and *TAGLN*). Notably, compared with the ANT tissues, we observed higher proportions of CapECs and PCs in tumour tissues (Extended Data Fig. 2b,c). Moreover, two tissue-specific ECs of renal glomerular cells (RGCs) (marked by *EMCN*, *SOST* and *PLAT*) and liver sinusoidal ECs (LSECs) (marked by *CLEC4G* and *CLEC1B*) were recognized in corresponding ANT tissues (Fig. 1d).

We next performed unsupervised graph clustering analysis of VECs, LECs and MCs, and divided these cells into 29 VEC (V0–V28), 8 LEC (L0–L7) and 16 MC (M0–M15) subclusters (Fig. 1e and Supplementary Fig. 1). We further estimated the cancer type enrichment of these subclusters by calculating the relative score of observed cell numbers over expected cell numbers ($R_{o/e}$ score; Fig. 1f and Methods). Most subclusters exhibited a pan-tumour origin. However, certain VEC and LEC subclusters, positioned at the periphery of the uniform manifold approximation and projection (UMAP) representation, exhibited similarities in tissue origin, including gynaecologic-cancer-derived CapECs (V26), breast cancer (BC)-derived CapECs (V28) and skin-tumour-derived CapECs and LECs (V27 and L4) (Fig. 1e,f). Analysis of the marker genes of these peripheral subclusters revealed that gynaecologic-derived CapECs specifically expressed *CXCL1* and *CXCL3*, which were associated with the progression of endometrial and ovarian cancers[11]. BC-derived CapECs exhibited high expression of *PREX2* and *RASGRF2*, which could be enhanced in response to VEGF[12]. Moreover, skin-tumour-derived CapECs and LECs expressed the skin-related markers *PERP* and *KRT14*[13] (Extended Data Fig. 2d). These subclusters were not included in the subsequent pan-tumour analysis.

## Tumour angiogenesis sprouting from VenECs

Tumour angiogenesis involves the development of neovascularization from established vascular beds[2,3]. To investigate the molecular underpinnings, we recapitulated tumour angiogenesis based on transcriptomic profiling of 116,958 pan-tumour VECs, including 18,048 ArtECs, 61,400 CapECs and 37,510 VenECs (Fig. 2a and Supplementary Fig. 2). In contrast to ANT tissues, tumour tissues were infiltrated with more CapECs than VenECs and ArtECs (Fig. 2b). To capture the trajectory of tumour angiogenesis, we performed a diffusion component analysis[14], which inferred a differentiation trajectory from VenECs to ArtECs passing through CapECs (Fig. 2c). This result was reproduced

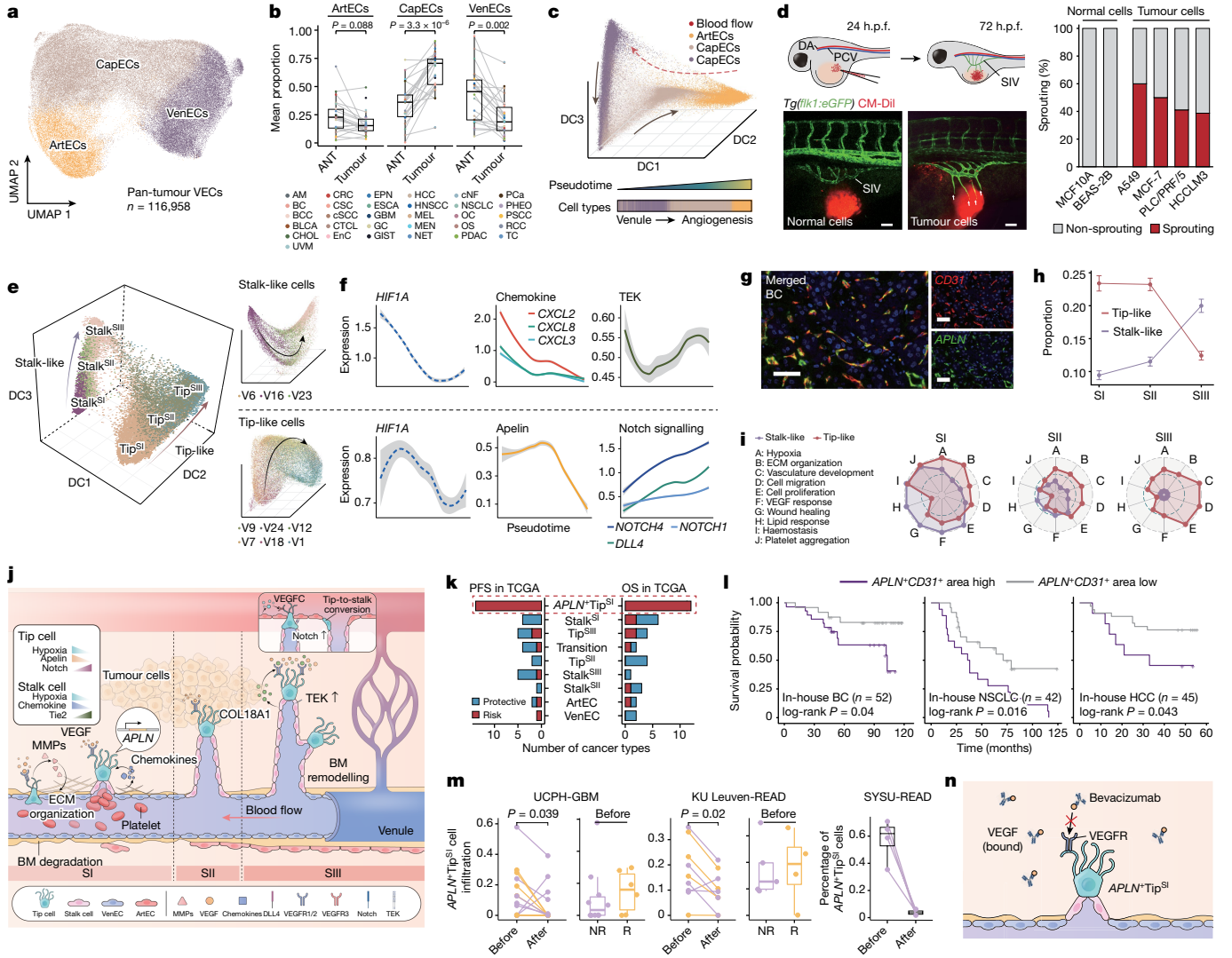


**Fig. 2 | Characterization of pan-tumour VECs. a**, UMAP analysis of pan-tumour VECs, coloured by inferred cell types. **b**, The mean proportions of each cancer type between ANT and tumour tissues across VEC subtypes. The dots represent cancer types ($n = 31$). **c**, Diffusion map indicating the trajectory of tumour angiogenesis. **d**, Tumour angiogenesis in transgenic *Tg(flk1:eGFP)* zebrafish embryos. Representative images display SIV morphology in zebrafish embryos xenotransplanted with normal cells (BEAS-2B) and tumour cells (A549), labelled with red fluorescent dye (CM-DiI). The bar plot shows the ectopic sprouting rates of the developing SIV of the zebrafish embryos. h.p.f., hours post-fertilization. **e**, Trajectories of tip-like and stalk-like cells in the diffusion map. **f**, Loess-regression-smoothened expression (*y* axis) of the indicated genes in pseudotime. The shaded region represents the 95% confidence intervals. **g**, Representative mIHC analysis of *APLN*$^+$ Tip$^{SI}$ cell markers in BC from $n = 10$ independent biological replicates. **h**, The proportion of angiogenic stages within tip-like and stalk-like cells (tip cells: $n = 9{,}063$ (SI), $n = 13{,}390$ (SII), $n = 6{,}351$ (SIII); stalk cells: $n = 4{,}399$ (SI), $n = 3{,}361$ (SII), $n = 5{,}517$ (SIII)). Data are mean ± s.e.m. **i**, Radar plot showing the scaled activities of TAPFs. **j**, Illustration of the putative process of tumour angiogenesis. **k**, The number of cancer types in which VECs are associated with overall survival/progression-free survival in TCGA cohorts. **l**, Kaplan–Meier curves of internal cohorts stratified by the low and high *APLN*$^+$*CD31*$^+$ area. **m**, Variations in *APLN*$^+$ Tip$^{SI}$ cell infiltration after bevacizumab treatment. The differences between pretreatment responders (R) and non-responders (NR) in UCPH-GBM ($n = 6$ (N), $n = 10$ (NR)), KU Leuven-READ ($n = 6$ (N), $n = 5$ (NR)) and SYSU-READ ($n = 4$) are shown. **n**, Illustration of the inhibition of VEGF-mediated angiogenesis by bevacizumab. For the box plots, the centre line shows the median, the box limits indicate the upper and lower quartiles, and the whiskers extend 1.5× the interquartile range. Statistical analysis was performed using two-sided Mann–Whitney *U*-tests (**b** and **m**) and log-rank tests (**l**). The diagram in **n** was created using BioRender. For **d** and **g**, scale bars, 50 µm.

using trajectory inference methods with different algorithmic principles (Methods and Extended Data Fig. 3a,b). Additional assessment of the cell differentiation ability[15] revealed that VenECs, during their transformation into CapECs, initially underwent a process of dedifferentiation (Extended Data Fig. 3c). We next constructed an in vivo tumour angiogenesis model by xenotransplanting tumour cells into zebrafish (Methods). We observed that tumour-induced neovascularization gradually extended from the subintestinal vein (SIV) towards the tumour location (Fig. 2d). However, this phenomenon did not occur when normal cells were injected into zebrafish. These results indicate that tumour angiogenesis was initiated from VenECs.

Neovascularization lacks the morphological characteristics of parental vasculature and can be interpreted as having capillary-like features[16]. Evaluation of cell purity also indicated that CapECs exhibited higher heterogeneity compared with VenECs and ArtECs (Extended Data Fig. 3d). During angiogenesis, tip cells sense and respond to guidance cues, leading the way, with stalk cells following behind to elongate the stalk of the bud and form a lumen[17]. We therefore scored the phenotypes of CapECs (Methods) and identified tip-like CapECs, stalk-like CapECs and CapECs in a transition state between tip and stalk cells (Extended Data Fig. 3e). We observed that tumour tissues were significantly enriched in tip-like CapECs (Extended Data Fig. 3f,g). Next, we traced

the differentiation trajectories of tip-like and stalk-like cells, defining the angiogenic stages as SI, SII and SIII (Extended Data Fig. 4a–c). Overall, the developmental trajectories of tip-like and stalk-like cells were independent and trended orthogonally in the 3D-diffusion component space (Fig. 2e). Both tip-like and stalk-like CapECs exhibited heightened activity of HIF-1α (encoded by *HIF1A*) at the SI stage (Extended Data Fig. 4d), followed by a gradual attenuation during SII and SIII (Fig. 2f), supporting hypoxia as having a pivotal role in triggering neoplastic sprouting.

As the angiogenic initiation, *APLN* (encoding apelin) was identified as the specific marker of $Tip^{SI}$ cells (hereafter $APLN^{+}$ $Tip^{SI}$ cells) (Fig. 2f and Extended Data Fig. 4e). Multicolour immunohistochemistry (mIHC) staining of tumour sections further confirmed the prevalence of $APLN^{+}$ $Tip^{SI}$ cells in various cancers (Fig. 2g and Extended Data Fig. 4f). Moreover, dynamic analysis of neovascularization showed that tip cells were predominantly found in the SI stage, whereas stalk cells were more abundant in the SIII stage (Fig. 2h). We next summarized the features of the tumour angiogenesis process (TAPFs) based on the functional analysis of overexpressed metagenes represented by a self-organizing map (SOM) analysis (Methods and Extended Data Fig. 4g). The result revealed ten major TAPFs, including hypoxia, extracellular matrix (ECM) organization, vascular development, cell migration, cell proliferation, VEGF response, wound healing, lipid response, haemostasis and platelet aggregation (Extended Data Fig. 4h). We next assessed active TAPFs using AUCell[18] and assigned them to neovascular cells (Fig. 2i). By performing further integration with modelling of intercellular communications (Methods, Extended Data Fig. 4i,j and Supplementary Table 3), we restored a putative process of dynamic tumour angiogenesis (Fig. 2j). During the SI stage, *APLN* and chemokines (*CXCL2*, *CXCL3* and *CXCL8*) were identified as the active ligands in tip and stalk cells, respectively (Extended Data Fig. 4i). $Tip^{SI}$ cells secrete apelin, which facilitates stalk cell proliferation and sprout elongation[19]. Simultaneously, chemokines released by $Stalk^{SI}$ cells can promote filopodia formation and migration of tip cells[20]. Both $APLN^{+}$ $Tip^{SI}$ and $Stalk^{SI}$ cells expressed ECM modulators to reorganize ECM and showed high platelet aggregation, haemostasis and wound-healing activities (Fig. 2i). By SIII, stalk cell proliferation decreased, whereas tip cell proliferation was enhanced (Fig. 2i). Moreover, we observed that the expression of angiopoietin receptor *TEK* in $Stalk^{SIII}$ cells was elevated (Extended Data Fig. 4e), which may contribute to stabilizing neovascularization[21–24]. *COL18A1* was also identified as a specific ligand in $Stalk^{SIII}$ cells (Extended Data Fig. 4i). *COL18A1*—a component of the vascular basement membrane[25]—can also bind to VEGF receptors (*FLT1* and *KDR*), resulting in competitive attenuation of VEGF signalling[26,27], implying basement membrane remodelling and angiogenesis completion during SIII. Moreover, Notch signalling was activated in $Tip^{SIII}$ cells (Fig. 2f), and activation of Notch signalling promoted $Tip^{SIII}$ cells to undergo tip-to-stalk phenotypic conversion through high expression of VEGFR3 (encoded by *FLT4*) and enabled the incorporation of tip cells into arterioles to limit angiogenesis[28–31] (Fig. 2j and Extended Data Fig. 4k).

To investigate the clinical significance of VECs, we assessed cell infiltration using deconvolutional analysis in The Cancer Genome Atlas (TCGA) cohorts (Methods and Supplementary Table 4). $APLN^{+}$ $Tip^{SI}$ cells were barely detectable in most ANT tissues (Extended Data Fig. 5a) but increased with higher tumour stages in TCGA tumour samples (Extended Data Fig. 5b). A Cox proportional-hazards model demonstrated that high infiltration of $APLN^{+}$ $Tip^{SI}$ cells predicted poor progression-free survival and overall survival (Fig. 2k and Extended Data Fig. 5c). Moreover, in our internal cohorts of BC, non-small cell lung cancer (NSCLC) and hepatocellular carcinoma (HCC), patients with tumours with a high proportion of $APLN^{+}$ $Tip^{SI}$ cells had poorer overall-survival outcomes (Fig. 2l and Supplementary Tables 5–7). We next examined the response of VEC subtypes to AAT. Bevacizumab—an anti-VEGF antibody—was developed to target tumour angiogenesis[32]. In three independent cohorts that were treated with bevacizumab combination therapy (Supplementary Table 1), $APLN^{+}$ $Tip^{SI}$ cell infiltration was substantially decreased after bevacizumab administration. Treatment responders also had higher levels of $APLN^{+}$ $Tip^{SI}$ cell infiltration before treatment compared with non-responders (Fig. 2m and Extended Data Fig. 5d,e), implying that $APLN^{+}$ $Tip^{SI}$ cells could be targeted for bevacizumab-induced normalization of tumour vessels (Fig. 2n). These findings further suggest that patients with high $APLN^{+}$ $Tip^{SI}$ cell tumour infiltration may benefit from AAT, notwithstanding their poorer prognosis.

## Differentiation lineages of tumour LECs

In total, 9,283 pan-tumour LECs (L0–L3 and L5–L7) were detected (Supplementary Fig. 3). We found that the L0 and L7 subclusters were mainly from the ANT samples (Extended Data Fig. 6a,b) and were therefore annotated as the normal-like LECs (NLECs). Further trajectory inference showed two distinct differentiation lineages originating from NLECs (Fig. 3a and Extended Data Fig. 6c). Trajectory 1 (T1) led to L2/L6 through L1/L5, and trajectory 2 (T2) differentiated directly to L3 through L1 (Extended Data Fig. 6d). We therefore characterized the properties of the L2/L6 and L3 lineages. Comparison of the marker expression levels for tip-like and stalk-like CapECs (Supplementary Table 2) between L2 and L6 cells indicated that L2 cells exhibit a tip-like phenotype, whereas L6 cells displayed a stalk-like phenotype (Fig. 3b). Gene expression analysis further revealed that tip-like (L2) and stalk-like (L6) LECs highly expressed *VEGFR3* and chemokines (*CCL2*, *CXCL1* and *CXCL2*), whereas L3 cells expressed multiple MHC-II molecules (*HLA-DP*, *HLA-DQ* and *HLA-DR*) and MHC-II-associated invariant chain (*CD74*) (Fig. 3c). We therefore annotated L3 as the antigen-presenting LEC (apLEC). Functional enrichment analysis also showed that tip-like and stalk-like LECs were associated with VEGF signalling and inflammatory responses and apLECs were significantly enriched in antigen presentation, immune processes and ECM organization (Extended Data Fig. 6e). We therefore inferred that T1 represented the process of tumour lymphangiogenesis (hereafter, T1-lymphangiogenesis), whereas T2 represented the differentiation of apLECs (hereafter, T2-apLEC). Notably, T1-lymphangiogenesis also exhibited higher *HIF1A* activity than T2-apLEC (Fig. 3c and Extended Data Fig. 6f), implying that hypoxia could be an inducer of tumour lymphangiogenesis.

We next identified lineage-related transcription factors (TFs) (Methods). Expression of the TF *PROX1* was significantly higher in T1-lymphangiogenesis than in T2-apLEC (Fig. 3d,e). Furthermore, *PROX1* was a downstream gene of the TF *NFKB1*, which was also enriched in T1-lymphangiogenesis. Functionally, both *PROX1* and *NFKB1* could be activated by inflammatory stimuli and regulated the expression of VEGFR3 in LECs, leading to lymphangiogenesis[33]. Contrary to T1-lymphangiogenesis, the anti-inflammatory TF *TSC22D3* was specifically expressed in apLECs (Fig. 3d,e and Extended Data Fig. 6g). Previous studies indicated that *TSC22D3* could inhibit NF-κB (encoded by *NFKB1*) nuclear translocation and DNA binding[34]. The evidence therefore suggests that the inflammatory response has a crucial role in LEC differentiation (Fig. 3f). Moreover, we determined cell differentiation status (Methods), which revealed a progressive increase in differentiation levels along T2-apLEC and an inverse decrease in differentiation levels along T1-lymphangiogenesis (Fig. 3e).

Notably, we observed a significant negative correlation between the proportion of apLECs and tip-like and stalk-like LECs (Extended Data Fig. 6h). The same trend was also observed in the TCGA cohorts. We confirmed the presence of apLECs ($TSC22D3^{high}NFKB1^{low}$) and tip-like LECs ($TSC22D3^{low}NFKB1^{high}$) in BC (Fig. 3g) and HCC (Extended Data Fig. 6i) using mIHC staining. In TCGA-HCC, high tumour infiltration of tip-like LECs was associated with poor prognostic outcomes, whereas apLEC infiltration could act as a protective factor (Extended Data Fig. 6j). The ratio of tip-like LEC to apLEC infiltration levels ($R_{T/ap}$) also emerged as an independent risk factor in the Cox regression

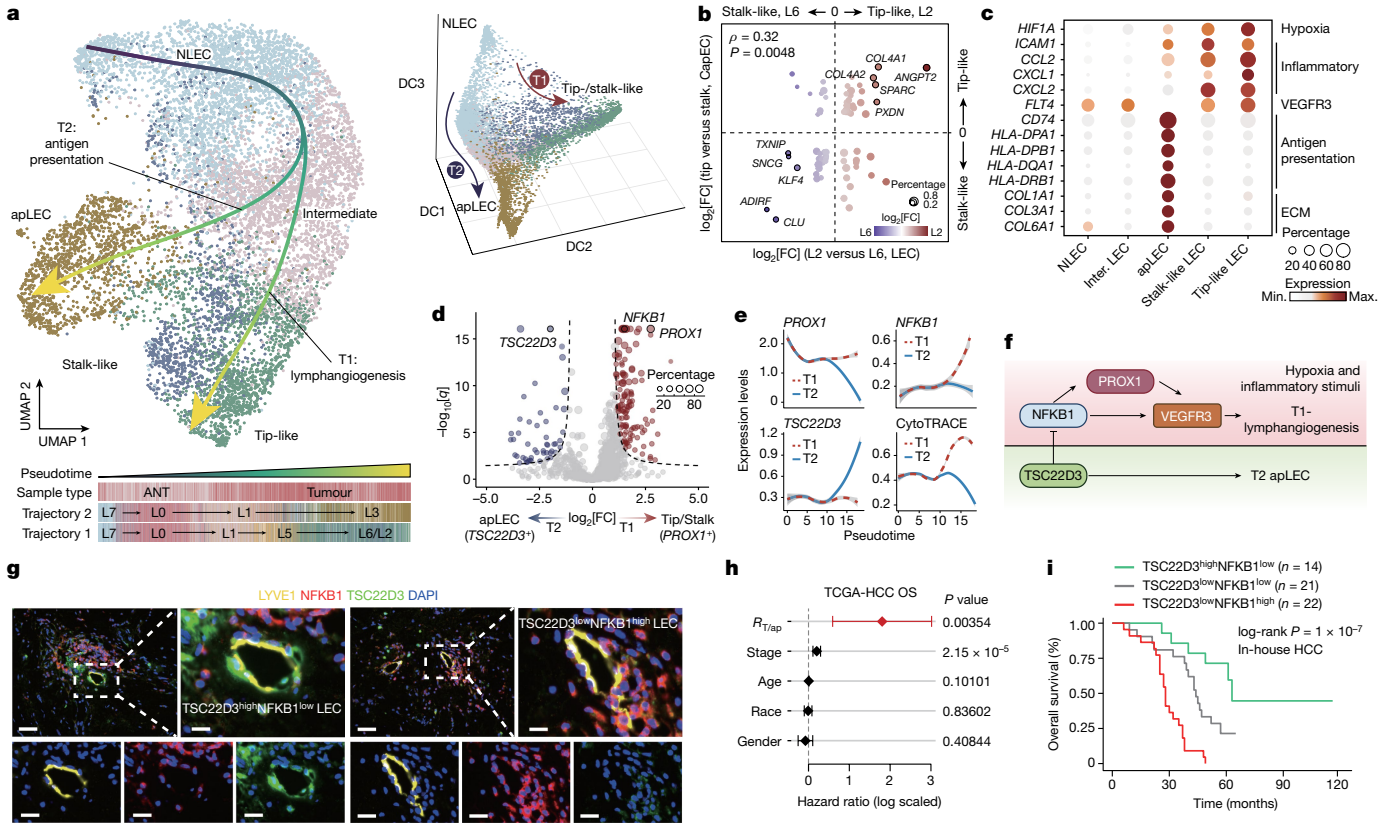


**Fig. 3 | Distinct LEC differentiation lineages. a**, The trajectories of LEC differentiation. A UMAP plot of Slingshot-derived LEC trajectories is shown on the left. The diffusion component map is shown on the right. Trajectory 1 (T1) and trajectory 2 (T2) originated from NLECs to tip-like/stalk-like LECs or apLECs, respectively. **b**, L2 highly expressed tip-like CapEC markers and L6 highly expressed stalk-like CapEC markers. The Spearman's correlation coefficient and its *P* value are shown. **c**, The expression levels of selected signature genes of tip-like/stalk-like LECs and apLECs. **d**, The differentially expressed TFs between T1 and T2. **e**, Loess-regression-smoothened expression of the indicated genes and CytoTRACE score (*y* axis) in pseudotime. The shaded region represents the 95% confidence intervals. **f**, The proposed TF regulation in LEC lineages. **g**, Representative mIHC analysis of markers of tip-like LECs (left) and apLECs (right) in BC from $n = 6$ independent biological replicates. Scale bars, 50 μm (top, columns 1 and 3), 20 μm (bottom) and 12.5 μm (top, columns 2 and 4). **h**, The hazard ratio (*x* axis, the centre of error bars) and 95% confidence intervals (error bars) associated with $R_{T/ap}$ (*y* axis) identified by multiple Cox proportional-hazards models with overall survival adjusted for tumour stage, age, race and gender in the TCGA-HCC cohort ($n = 363$ cases). Cox proportional hazard model two-sided *P* values are shown. **i**, Kaplan–Meier survival curves of patients divided into the indicated groups in the in-house HCC cohort. For **i**, statistical analysis was performed using log-rank tests.

models with corrections made for age, gender, race and tumour stage (Fig. 3h). Consistently, we observed that patients with a high proportion of apLECs (*TSC22D3*$^{high}$*NFKB1*$^{low}$) had longer overall survival outcomes compared with those who had a high proportion of tip-like LECs (*TSC22D3*$^{low}$*NFKB1*$^{high}$) in our internal HCC cohort (Fig. 3i and Supplementary Table 8). The phenomena also presented in multiple cancers (Extended Data Fig. 6k), implying that distinct LEC differentiation lineages may be linked to varying prognostic outcomes.

## ER stress drove proangiogenic matPCs

We detected 40,391 MCs in the TVM, including 29,572 PCs and 10,819 SMCs (Supplementary Fig. 4). Compared with ANT tissues, the proportion of PCs was significantly elevated compared with that of SMCs in tumour tissues (Extended Data Fig. 7a). We also noted lower cell purity in PCs compared with SMCs (Extended Data Fig. 7b), prompting the heterogeneous phenotypes of PCs. Four PC subtypes were identified across pan-tumour tissues (Extended Data Fig. 7c): (1) vasodilation-associated PCs (vdPCs); (2) adipocyte-like PCs; (3) myoid-like PCs; and (4) matrix-producing PCs (matPCs) (Fig. 4a). Notably, the guanylate cyclase subunits (*GUCY1A3* and *GUCY1B3*) expressed by vdPCs were also related to the quiescent PC signature[35]. We therefore scored the quiescent states of PCs on the basis of the quiescent PC signature (Methods). vdPCs were characterized by higher quiescent scores than other PC subtypes (Extended Data Fig. 7d). We also analysed the differentially expressed genes (DEGs) between quiescent high ($Q^{high}$) and quiescent low ($Q^{low}$) cells, which showed that $Q^{high}$ cells highly expressed oxidative-phosphorylation-related genes and $Q^{low}$ cells highly expressed ECM and glycolysis regulators (Fig. 4b), implying that metabolic reprogramming is closely associated with PC activity[36–38].

Notably, matPCs showed quiescence and metabolic state transitioning (Extended Data Fig. 7d–f). We next ordered matPCs along pseudotime and observed that matPCs in the $Q^{high}$ state (matPC$^{Q}$) could differentiate into *BASP1*$^{+}$ matPCs (Fig. 4c and Extended Data Fig. 7g,h). Further activity analysis demonstrated that *BASP1*$^{+}$ matPCs presented with lower tight junction, oxidative phosphorylation and quiescence levels and higher angiogenic, hypoxia and glycolysis levels compared with matPC$^{Q}$ cells (Fig. 4c and Extended Data Fig. 7i). Notably, *BASP1*$^{+}$ matPCs also expressed osteopontin (encoded by *SPP1*) (Fig. 4c). Perivascular osteopontin has been reported to have proangiogenic effects[39]. We further validated the presence of *BASP1*$^{+}$ matPCs in glioblastoma (GBM) using mIHC staining (Fig. 4d). We next evaluated TF activity of matPCs[18]. Endoplasmic reticulum (ER)-stress-related TFs *ATF3* and *XBP1*[40,41] showed a clear association with pseudotime at both the transcription and TF activity levels (Fig. 4e and Extended Data Fig. 7j). Our analysis of ER-stress-related genes also indicated that ER stress was provoked during matPC differentiation (Extended Data Fig. 7k).

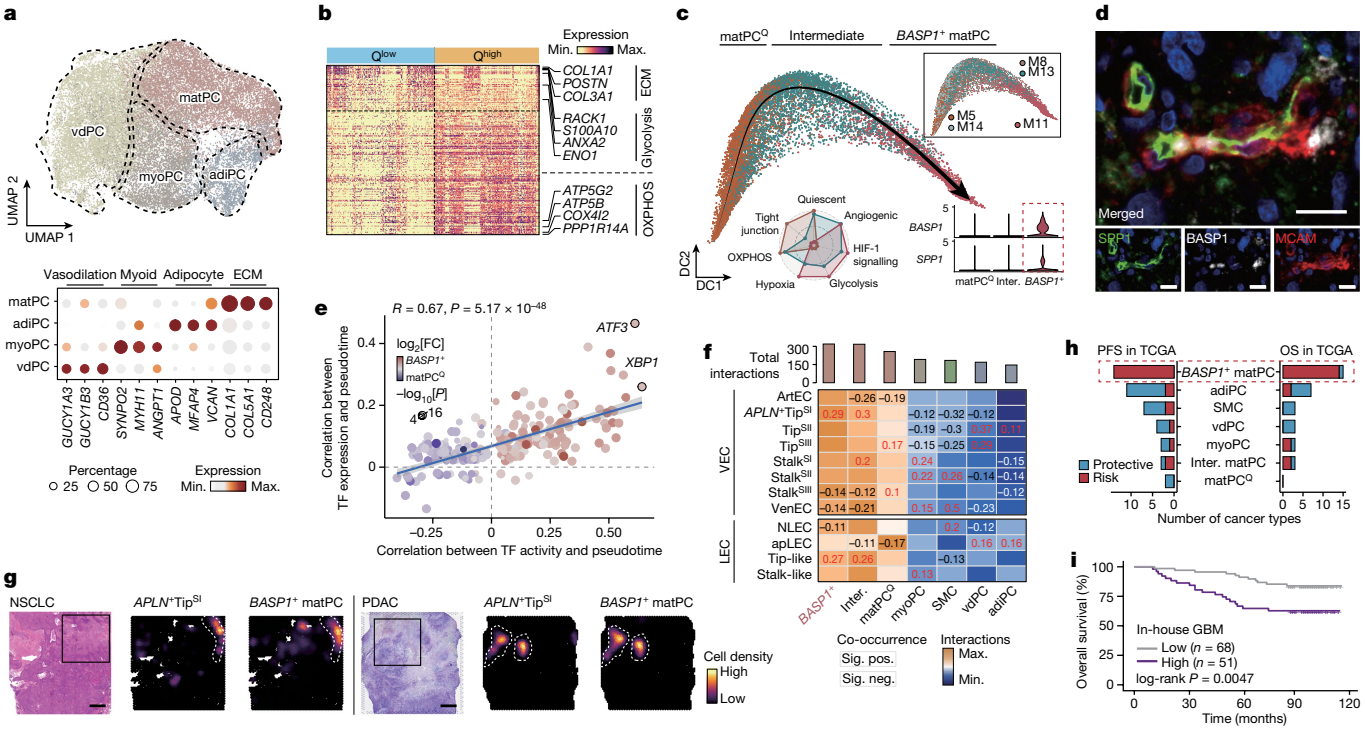


**Fig. 4 | Characterization of pan-tumour MCs. a**, UMAP analysis of PCs, coloured by PC phenotypes. The expression of selected genes for PC phenotypes is shown at the bottom. **b**, DEGs between $Q^{high}$ and $Q^{low}$ PCs. **c**, The trajectory of matPC differentiation. The radar plot shows the scaled activities of PC attributes. The violin plot shows the expression of *BASP1* and *SPP1* across matPC subtypes. **d**, Representative mIHC analysis of *BASP1*$^+$ matPC markers in GBM from $n = 6$ independent biological replicates. Scale bars, 10 µm. **e**, Pseudotime-related TFs. Spearman's correlation coefficients were calculated between pseudotime and TF transcription (*y* axis) and TF SCENIC score (*x* axis), respectively. The Spearman's correlation coefficient between the *x* axis and *y* axis and its *P* value are shown. Dots represent TFs, coloured by the fold change and sized according to the −log[*P*] from a two-sided Mann–Whitney *U*-test between matPC$^Q$ and *BASP1*$^+$ matPC. **f**, Cell–cell interactions between MCs and ECs. The bar plot shows the total number of LR interactions of MCs. The heat map shows the number of interactions between MC subtype and EC subtype. Significant co-occurring Pearson's correlation coefficients are indicated on the heat map; positive correlations (red) and negative correlations (black) are indicated. **g**, Co-localization between *APLN*$^+$ Tip$^{SI}$ cells and *BASP1*$^+$ matPCs at spatial positions from spatial transcriptomics data of NSCLC ($n = 1$ case) and pancreatic carcinoma ($n = 1$ case). Scale bars, 1 mm. **h**, The number of cancer types in which MCs exhibited a clear correlation with overall survival and progression-free survival in the TCGA cohorts. **i**, Kaplan–Meier survival curves of patients with low and high proportions of *BASP1*$^+$ matPCs in the in-house GBM cohort. For **i**, statistical analysis was performed using log-rank tests.

Moreover, several ER-stress-associated TFs exhibited high activities in *BASP1*$^+$ matPCs (Extended Data Fig. 7l,m). These results demonstrate that ER stress may be a key factor in the evolution of matPCs.

We subsequently analysed the intercellular interactions between MCs and ECs (Methods and Supplementary Table 9). *BASP1*$^+$ matPCs showed the highest interaction frequency among all MC subtypes, highlighting their high activity in the TVM (Fig. 4f). Notably, a significant positive association was observed between the proportion of *BASP1*$^+$ matPCs and *APLN*$^+$ Tip$^{SI}$ cells (Extended Data Fig. 8a). We next performed spatial transcriptomics sequencing on a haematoxylin and eosin section of NSCLC (Methods). In both internal and external spatial transcriptomics data (Supplementary Table 1), we performed spatial transcriptomics deconvolution analyses[42]. The results showed that *BASP1*$^+$ matPCs and *APLN*$^+$ Tip$^{SI}$ cells were spatially proximate (Fig. 4g and Extended Data Fig. 8b). Ligand–receptor (LR) interaction indicated that *BASP1*$^+$ matPCs could secrete VEGF, facilitating angiogenesis (Extended Data Fig. 8c,d). Chromatin immunoprecipitation followed by sequencing (ChIP–seq) data showed that *VEGFA* and *VEGFB* were also potential downstream genes of the TFs XBP1 and ATF3 (Extended Data Fig. 8e), supporting a critical role for ER stress in tumour angiogenesis[40]. Furthermore, tumour-infiltrated *BASP1*$^+$ matPCs were not only elevated in samples of higher tumour stages (Extended Data Fig. 8f) but also associated with disease progression and poor prognosis in TCGA cohorts (Fig. 4h and Extended Data Fig. 8g). We also validated that a high proportion of *BASP1*$^+$ matPCs was associated with worse overall survival in the in-house GBM cohort (Fig. 4i and Supplementary Table 10).

## Cross-talk between vasculature and the TME

To dissect vasculature-associated intercellular communications, we sorted 160,366 myeloid cells, 261,751 T cells and natural killer cells (T/NK cells), 42,342 B cells and 92,297 fibroblasts and determined their subtypes (Methods and Supplementary Fig. 5). We further identified homogeneous metacells in the tumour microenvironment (TME) (Fig. 5a) and performed an unbiased LR interaction analysis between vascular cells and TME cells (Methods and Supplementary Table 11). Myeloid cells and fibroblasts showed more interactions than T/NK and B cells (Fig. 5b), including the pivotal VEGF signalling. Analysis of VEGF–VEGFR signalling (VEGFA, VEGFB–FLT1, KDR) indicated that *SPP1*$^+$ tumour-associated macrophages (TAMs), myeloid-derived suppressor cells (MDSCs), monocytes, type 2 conventional dendritic cells (cDC2s), cancer-associated myofibroblasts (myoCAFs), matrix-associated CAFs (matCAFs) and *BASP1*$^+$ matPCs were the main sources of VEGF for neovascularization (Extended Data Figs. 8c and 9a,b). We observed that *SPP1*$^+$ TAMs showed the highest interaction numbers with vascular cells (Fig. 5b). *SPP1*$^+$ TAMs also significantly co-occurred with tip cells (Extended Data Fig. 9c), and the interaction numbers with *APLN*$^+$ Tip$^{SI}$ cells were higher than those with SII and SIII tip cells (Extended Data Fig. 9d). Spatial transcriptomics data suggested spatial co-localization of *APLN*$^+$ Tip$^{SI}$ cells and *SPP1*$^+$ TAMs (Fig. 5c and Extended Data Fig. 9e). Moreover, mIHC analysis provided additional evidence supporting the proximity of *SPP1*$^+$*CD68*$^+$ TAMs and *APLN*$^+$*CD31*$^+$ ECs in situ (Fig. 5d and Extended Data Fig. 9f).

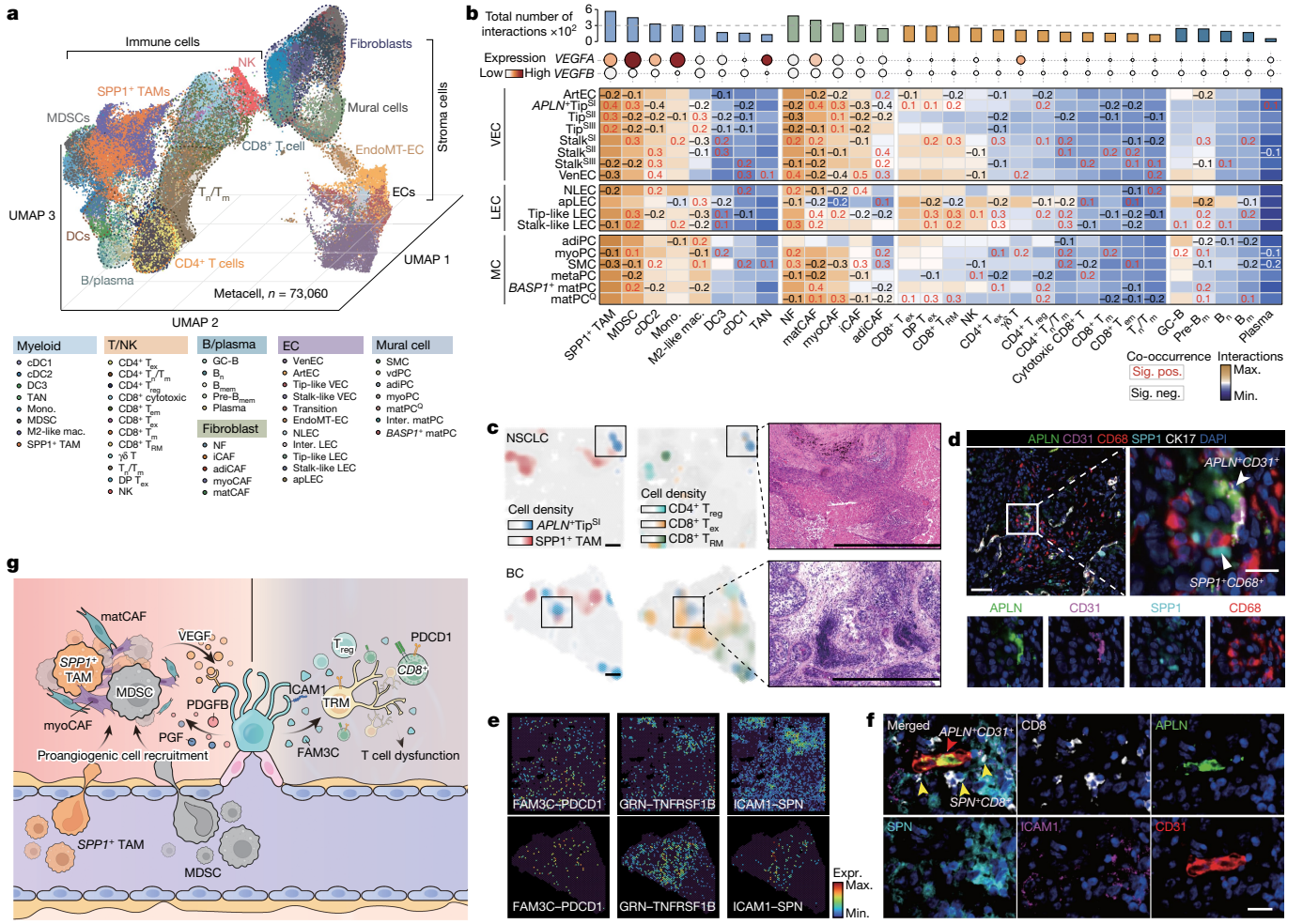


**Fig. 5 | Cross-talk between vasculature and TME cells. a**, UMAP of pan-tumour TME and vascular metacells, coloured by annotated cell identity. The dots represent the metacells produced by SuperCell. $B_{mem}$, memory B cells; DP, $CD4^+CD8^+$ double positive; GC-B, germinal centre B cells; Mac., macrophages; Mono., monocytes; Pre-$B_{mem}$, precursor memory B cells; $T_{EM}$, effector memory T cells; $T_{exh}$, exhausted T cells; $T_n/T_m$, naive and memory T cells; $T_{reg}$, regulatory T cells; $T_{RM}$, resident memory T cells; TAN, tumor-associated neutrophils; NF, normal-like fibroblasts. **b**, Cell–cell interactions between TME cells and vascular cells. The bar plot shows the number of total LR interactions for TME cells. The heat map shows the number of LR interactions between vascular cells and TME cells. Significant co-occurring Pearson's correlation coefficients are indicated on the heat map; significant positive (sig. pos.) correlations (red) and significant negative (sig. neg.) correlations (black) are shown. **c**, Colocalization between *APLN*$^+$ $Tip^{SI}$ cells, *SPP1*$^+$ TAMs and exhausted T cells in spatial transcriptomics data of NSCLC ($n = 1$ case) and BC ($n = 1$ case). Localized magnified haematoxylin and eosin image. Scale bars, 1 mm. **d**, Representative mIHC analysis of *APLN*$^+$ tip cells and *SPP1*$^+$ TAMs in BC from $n = 6$ independent biological replicates. The arrows depict the specific cell types. Scale bars, 50 µm (left) and 15 µm (right). **e**, Spatial plot showing the expression (expr.) intensity of LR pairs for immunosuppressive interactions. Expression intensity was calculated by the geometric mean of an LR pair. **f**, Representative mIHC of *ICAM1*-expressing *APLN*$^+$ tip cells and *SPN*-expressing *CD8*$^+$ T cells in NSCLC from $n = 5$ independent biological replicates. The arrows depict the specific cell types. Scale bar, 20 µm. **g**, Illustration showing that *APLN*$^+$ $Tip^{SI}$ cells shape a proangiogenic immunosuppressive microenvironment. The diagram in **g** was created using BioRender.

In addition to *SPP1*$^+$ TAMs and *BASP1*$^+$ matPCs, we found other proangiogenic cells that significantly co-occurred with *APLN*$^+$ $Tip^{SI}$ cells, including MDSCs, myoCAFs and matCAFs. LR analysis indicated that *APLN*$^+$ $Tip^{SI}$ cells could not only release the chemoattractant *PDGFB* and *PGF* to recruit stromal proangiogenic cells (*BASP1*$^+$ matPCs, myoCAFs and matCAFs) but also facilitated the extravasation of myeloid proangiogenic cells (*SPP1*$^+$ TAMs and MDSCs) to tumour tissue through CD99–PILRA[43] (Extended Data Figs. 8c, 9a and 10a). Moreover, mIHC staining revealed the co-localization of *PDGFB*-expressing *APLN*$^+$ tip cells and *PDGFRB*-expressing myoCAFs (aSMA$^+$) (Extended Data Fig. 10b). These proangiogenic cells were also physically close to each other, forming a supporting niche (Extended Data Fig. 9e). Notably, *APLN*$^+$ $Tip^{SI}$ cells showed immunomodulatory potential for T cells. Regulatory T ($T_{reg}$) cells, tissue-resident memory T cells and exhausted T cells co-occurred with *APLN*$^+$ $Tip^{SI}$ cells (Fig. 5c and Extended Data Fig. 9e). The ligands PODXL and CD34, which are expressed by *APLN*$^+$ $Tip^{SI}$ cells, could interact with their homing receptor L-selectin (encoded by *SELL*) on $T_{reg}$ cells (Extended Data Fig. 10c), facilitating $T_{reg}$ cell tumour infiltration[44]. Moreover, *APLN*$^+$ $Tip^{SI}$ cells could induce T cell dysfunction and exhaustion through FAM3C–PDCD1, GRN–TNFRSF1B and ICAM1–SPN pairs (Fig. 5e). mIHC staining confirmed the colocalization of *ICAM1*-expressing *APLN*$^+$ tip cells and *SPN*-expressing *CD8*$^+$ T cells in NSCLC (Fig. 5f), which could be associated with cytotoxic T cell inhibition[45]. This immunosuppressive effect also persisted in the SII/SIII neovascularization and lymphatic vessels (Extended Data Fig. 10d). These results suggest that *APLN*$^+$ $Tip^{SI}$ cells contribute to shaping a proangiogenic immunosuppressive TME (Fig. 5g).

## Discussion

Tumour angiogenesis is a vital process for tumours, but the origin of the ECs that propel this dynamic process remains an important

unresolved issue. Here we demonstrate that VenECs are the starting point of angiogenic initiation using in silico and in vivo models, which further provide compelling evidence that tumour angiogenesis originates from venules[28,29,46–49] (Supplementary Discussion). After initiation of the tumour angiogenic switch, *APLN*$^{+}$ Tip$^{SI}$ cell sprouts emerge from the venous endothelium under the influence of hypoxia and VEGF signalling. Furthermore, while *APLN*$^{+}$ Tip$^{SI}$ cell presence in the tumour tissue can be used as a prognostic indicator for poor outcomes and disease progression across various cancer types, it presents a potential target for bevacizumab-mediated angiogenesis disruption. This evidence also raises the question of whether increased *APLN*$^{+}$ Tip$^{SI}$ cell vessel counts in tumour biopsies could be considered to be a predictive biomarker of clinical benefit in future prospective randomized trials of bevacizumab. Lymphangiogenesis was activated by inflammatory and hypoxia TVM[33]. Notably, we found a distinct LEC lineage, apLEC, which specifically expresses the anti-inflammatory TF *TSC22D3*. Recent studies showed that TSC22D3 can inhibit NF-κB DNA binding[34] and can mediate the ubiquitin-dependent degradation of HIF-1α[50]. We also analysed MC taxonomy, emphasizing the heterogeneous matPCs. One such pro-angiogenic matPC, characterized by *BASP1* expression[35], was driven by ER and oxidative stress. This matPC exhibited reduced tight-junction levels, facilitating tip cell sprouting[51]. Tumour-infiltrated *BASP1*$^{+}$ matPCs were associated with poor prognosis in multiple cancers. In summary, our comprehensive vascular compendium will facilitate exploration of tumour vascular biology and provide computational sources for potential targets of clinical significance and AAT.

## Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-024-07698-1.


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# Methods

This study complied with all relevant ethical regulations and was approved by the Ethics Committee of Chongqing University Three Gorges Hospital. All of the participants provided written informed consent.

### Single-cell library preparation and sequencing

The BD Rhapsody system, Singleron platform and SeekOne DD Chip were used in our study.

**Sequencing using the BD Rhapsody system.** The cells from each patient were labelled with sample tags from the Human Immune Single-Cell Multiplexing Kit (BD Biosciences) according to the manufacturer's protocol. The cell capture beads were retrieved for reverse transcription according to the manufacturer's protocol (BD Biosciences, Single-Cell Capture and cDNA Synthesis). Single-cell capture and cDNA library preparation were performed using the BD Rhapsody Express Single-Cell Analysis System (BD Biosciences) according to the manufacturer's protocol. The libraries were loaded onto an S1 flow cell (2 × 100 cycle) and paired-end sequenced at >200,000 reads per cell depth using the NovaSeq 6000 Sequencer (Illumina) at the Kinghorn Centre for Cellular Genomics, Garvan Institute. PhiX (20%) was added to the sequencing run to compensate for the low-complexity library[52].

**Sequencing using the Singleron platform.** Single-cell suspensions ($1 \times 10^5$ cells per ml) with PBS (HyClone) were loaded into microfluidic devices using the Singleron Matrix Single-Cell Processing System (Singleron). Next, the scRNA-seq libraries were constructed according to the protocol of the GEXSCOPE single-cell RNA library kits (Singleron)[53]. Individual libraries were diluted to 4 nM and pooled for sequencing. Finally, the pools were sequenced on the Illumina HiSeq X system with 150 bp paired-end reads.

**Sequencing with the SeekOne DD chip.** scRNA-seq libraries were prepared using SeekOne Digital Droplet Single-Cell 3′ library preparation kit (SeekGene). In brief, the appropriate number of cells were mixed with reverse transcription reagent and then added to the sample well in the SeekOne DD Chip S3. Subsequently, barcoded hydrogel beads and partitioning oil were dispensed into corresponding wells separately in the chip S3. Emulsion droplet generation reverse transcription was performed at 42 °C for 90 min and inactivated at 80 °C for 15 min. Next, cDNA was purified from the broken droplet and amplified in a PCR reaction. The amplified cDNA product was then cleaned, fragmented, end-repaired, A-tailed and ligated to the sequencing adaptor. Finally, the indexed PCR was performed to amplify the DNA representing 3′ poly(A) part of expressing genes, which also contained a cell barcode and unique molecular index. The indexed sequencing libraries were cleaned up with SPRI beads, quantified by quantitative PCR (KAPA Biosystems, KK4824), and then sequenced on the Illumina NovaSeq 6000 system with paired-end 150 bp read length or the DNBSEQ-T7 platform with paired-end 100 bp read length[54,55].

### Spatial transcriptomics preparation and sequencing

Spatial transcriptomic profiling was conducted using the 10x Genomics Visium CytAssist platform. Initially, a single formalin-fixed paraffin-embedded (FFPE) tissue sample was obtained from a patient diagnosed with lung adenocarcinoma. The tissue was carefully mounted onto Visium CytAssist Gene Expression Slides and underwent a series of procedures, including deparaffinization, staining, imaging, decrosslinking and probe hybridization, with the aim of capturing spatially resolved mRNA transcripts. Subsequently, the Visium CytAssist gene expression chips were inserted into the Visium CytAssist instrument for bright-field imaging. Within the instrument, analyte transfer was performed, allowing precise capture of transcriptomic information from the FFPE tissue. Raw sequencing data were processed using the Space Ranger pipeline (v.2.1.1).

### mIHC analysis

Human tissue samples were provided by Chongqing University Three Gorges Hospital. The specimens were collected within 30 min after tumour resection and fixed in formalin for 48 h. Dehydration and embedding in paraffin were performed according to routine methods. Paraffin blocks were cut into 5 µm slides and adhered onto the glass slides.

For mIHC staining, 5-µm-thick FFPE sections of BC tissues were stained with the Opal 7 colour fluorescent IHC Kit (PerkinElmer). First, deparaffinization, rehydration and permeabilization were performed on all slides, followed by 15 min of Tris-EDTA antigen retrieval under high temperature and high pressure. Next, the slides were incubated with primary antibodies, secondary-HRP antibodies and Opal TSA dyes for 16 h (4 °C), 10 min (room temperature) and 20 min (room temperature), respectively. Subsequent rounds of staining consisted of antigen retrieval, primary antibodies, secondary-HRP antibodies and Opal TSA dyes. The primary antibodies were as follows: CD31 (Abcam, ab28364, 1:100), APLN (Abcam, ab230536, 1:200), CK17 (Abcam, ab53707, 1:100), SPP1 (R&D Systems, AF1433, 1:100), CD68 (Abcam, ab213363, 1:8,000), MCAM (CST, 81701T, 1:200), LYVE1 (Abcam, ab219556, 1:1,000), NFKB1 (Abcam, ab305263, 1:500), TSC22D3 (Thermo Fisher Scientific, PA5-98816, 1:500), BASP1 (Abcam, ab312322, 1:2,000), PDGFB (Abcam, ab107101, 1:1,000), PDGFRB (Abcam, ab313777, 1:500), ICAM1 (Abcam, ab109361, 1:500), SPN (Abcam, ab235452, 1:2,000) and aSMA (CST, 19245T, 1:1,000). The primary antibodies were as follows: 1× Opal Anti-Ms + Rb HRP or HRP-conjugated rabbit anti-goat IgG (H+L) (ABclonal, AS029, 1:1,000). DAPI was used for nuclear counterstaining. The slides were finally mounted with antifade reagent (G1401, Servicebio). PhenoImager HT was used to capture the images and identify all markers of interest. Three to seven representative fields of the whole-slide scan images were selected and quantitatively analysed using ImageJ. The staining and analysis results of the mIHC were also checked by two certified pathologists.

### Cell injection and xenograft zebrafish models

Zebrafish eggs were incubated at 28.5 °C in 0.3× Danieau's solution (19.3 mM NaCl, 0.23 mM KCl, 0.13 mM $MgSO_4$, 0.2 mM $Ca(NO_3)_2$, 1.7 mM HEPES, pH 7.0). To inhibit pigmentation, 1-phenyl-2-thiourea (30 µg $ml^{-1}$) was added to the medium at 20 h after fertilization. Transgenic *Tg(flk1:eGFP)* zebrafish embryos, screened for green fluorescent vasculature, were manually dechorionated using two needles at 24 h after fertilization[56]. The embryos were anaesthetized with tricaine (50 µg $ml^{-1}$) and transferred onto an agar mould for microinjection. A549, MCF7, PLC/PRF/5, BEAS-2B and MCF10A cells were purchased from American Type Culture Collection (ATCC). HCCLM3 cells were purchased from the China Center for Type Culture Collection (CCTCC). All cell lines were routinely tested and were negative for mycoplasma. Both the cancer cell lines (A549, MCF-7, PLC/PRF/5, HCCLM3) and the normal human cell lines (BEAS-2B and MCF10A) were labelled with 5 µM of chloromethylbenzamido-DiI (CM-DiI) cell tracker dye and then injected into a superficial location of the yolk near the perivitelline space of the embryos at 24 h after fertilization using a microinjector. Approximately 300–500 cells were injected into each embryo. After transplantation, the embryos were immediately placed at 31 °C for recovery. An hour later, embryos with cells injected off the transplantation site were discarded, and only embryos with the same amount of fluorescent cells near the future SIV region were kept at 28.5 °C to continue development. At 72 h after fertilization, embryos with injected cells were imaged using a ×25 water objective on the Leica SP8 with a Bio-Rad 1024 laser-scanning confocal imaging system equipped with a krypton–argon ion laser. All experimental procedures using zebrafish/tumour xenograft research were approved by the Ethics Committee of Chongqing University Three Gorges Hospital.

### Single-cell raw data processing

In addition to the data generated in this study, we collected 58 publicly available external single-cell datasets (Supplementary Table 1). For the original scRNA-seq data available, we quantified gene expression count matrices; otherwise, the CPM/TPM matrices were downloaded and used for downstream analysis. The datasets generated by the four techniques (10x Genomics Chromium, BD Rhapsody, Singleron and SeekOne DD Chip platforms) were processed as follows. For data generated using the 10x platform, the raw reads were processed using the CellRanger toolkit (v.6.1.2). For data generated with the BD Rhapsody platform, the raw data were processed according to the BD Rhapsody Sequence Analysis Pipeline (BD Biosciences). For data generated using the Singleron platform, the raw reads were processed using the scopetools (https://github.com/SingleronBio/SCOPE-tools). For data generated using the SeekOne DD Chip platform, the raw reads were processed using the SeekOneTools (Beijing SeekGene BioSciences). All reads were aligned against the reference genome GRCh38 (ensemble version 99).

### Quality control, identification of vascular cells, data integration and batch correction

**In silico sorting of high-quality vascular cells.** First, we applied the Seurat pipeline to each single-cell dataset (Supplementary Table 1) to perform in silico sorting of high-quality vascular cells[57]. We next calculated the percentages of mitochondrial and ribosomal UMI counts and the S-phase score and G2M-phase score using the CellCycleScoring function. We then filtered out cells with >30% mitochondrial UMI counts and used the Scrublet with the default parameters to remove potential doublets[58]. The gene count expression matrix was normalized using the LogNormalize method with a scale factor 10,000, and the top 2,000 highly variable genes (HVGs) were identified with the vst method using the FindVariableFeatures function. Subsequently, we normalized the HVG expression matrix using the SCTransform function to regress out the effects of percentages of mitochondrial and ribosomal genes, cell cycle and donors. Principal component analysis was performed using the RunPCA function. The top 30 principal components were used to correct the effects of techniques and samples using the RunHarmony function in the R package harmony[59]. We constructed an SNN graph using the FindNeighbors function and clustered cells using the Louvain algorithm with the FindClusters function based on the harmony reduction. For all datasets, the resolution parameter was set to 2.0, allowing sufficient fine clusters to be produced[60]. Next, we annotated vascular cell clusters according to the expression of canonical markers: *PECAM1* for ECs and *RGS5*, *PDGFRB*, *ACTA2* and *TAGLN* for MCs. We identified the DEGs in each vasculature cluster using the FindAllMarkers function. Some clusters were found to highly express genes encoding heat shock proteins, which was probably related to the previously described tissue dissociation operation[60–62]. We counted marker genes with recurrence rates of more than 30% across dissociation-related clusters, which were defined as disassociation-induced genes. The AddModuleScore function was used to compute the DIG score. Some contaminating clusters, which showed the signatures of melanocytes (*PMEL*, *MLANA* and *TYRP1*), epithelial cells (*EPCAM*), red blood cells (*HBB*, *HBA1* and *HBA2*), myeloid cells (*CD163*, *CD14* and *LYZ*), T/NK cells (*CD2*, *CD3D*, *PTPRC* and *KLRD1*) and B/plasma cells (*CD79A* and *MS4A1*), were recognized. These dissociation-related and contaminating clusters were excluded from downstream integration.

**Data integration and batch correction.** We next integrated the heterogeneous vasculature cells produced by multiple platforms. Genes expressed by less than 100 cells and cells expressing less than 200 genes or more than 4,000 genes were removed. We normalized the integrated gene expression matrix and detected the top 2,000 HVGs. To reduce the effects of percentages of mitochondrial and ribosomal genes, cell cycle, DIG signature, techniques and donors, we scaled the HVG count matrix to *z*-score using the ScaleData function. To remove batch effects, we calculated cluster similarity spectrum (CSS) using the function cluster_sim_spectrum in the R package simspec with the default parameters (label_tag = "DonorID", spectrum_type = "corr_ztransform", corr_method = "spearman")[63]. In brief, the cells from each donor were subgrouped, and Louvain clustering with a resolution of 0.6 was applied based on the precalculated top 20 principal components. The average expression of the HVGs was calculated for each cluster in each donor. The Spearman's correlation coefficient between cells and clusters in each donor was then computed. For each cell, its correlations with different clusters of each donor were *z*-transformed (similarity spectrum), which were concatenated as the final CSS representation[64,65]. UMAP was performed on the CSS representation (Extended Data Fig. 1a,b). We also quantified the integration by calculating the local inverse Simpson's index (LISI) using the compute_lisi function in the R package LISI (https://github.com/immunogenomics/LISI), which reflected the effective number of datasets in a cell's neighbourhood (Extended Data Fig. 1c). Higher LISI values indicated that neighbourhoods were supplied by more datasets, resulting in a lower batch effect[59].

### Vascular cell subclustering

We built a shared nearest neighbour (SNN) map and used the Louvain algorithm to calculate the granularity of the clustering for LECs, VECs and MCs using Seurat function FindClusters. We generated 29 VECs (V0–V28), eight LECs (L0–L7) and 16 MCs (M0–M15) subclusters (Fig. 1f) and identified subcluster-related DEGs using the function FindAllMarkers with the default parameters (Supplementary Fig. 1). These subclusters were subsequently used to test cancer/tissue type enrichment and cell subtype identification.

### Cancer type enrichment analysis

Previous studies showed that certain types of endothelial and stromal cells were typically tissue specific[7,66]. We therefore calculated the ratio of the observed cell number over the expected cell number of a given vascular cell cluster and cancer type to identify the shared vascular cells in pan-cancer according to the formula $R_{o/e}$ = observed/expected, where the expected cell numbers for each combination of cell clusters and cancer types were obtained from the $\chi^2$ test[67]. If $R_{o/e} > 1$, we considered that the cells from a given cell cluster were observed more frequently than random expectations in the specific cancer type (enriched); otherwise, they were observed less frequently than random expectations in the specific cancer type (depleted) (Fig. 1e,f).

### Characterization of CapEC phenotypes

We first collected gene signatures of the vascular tip and stalk cells[68–70]. Curated tip and stalk cell signatures are as follows: Tip cell: *ADM*, *ANGPT2*, *ANKRD37*, *APLN*, *C1QTNF6*, *CD93*, *CLDN5*, *COL4A1*, *COL4A2*, *COTL1*, *CXCR4*, *DLL4*, *EDNRB*, *ESM1*, *FSCN1*, *GPIHBP1*, *HSPG2*, *IGFBP3*, *INHBB*, *ITGA5*, *JUP*, *KCNE3*, *KCNJ8*, *KDR*, *LAMA4*, *LAMB1*, *LAMC1*, *LXN*, *MARCKS*, *MARCKSL1*, *MCAM*, *MEST*, *MYH9*, *MYO1B*, *N4BP3*, *NID2*, *NOTCH4*, *PDGFB*, *PGF*, *PLOD1*, *PLXND1*, *PMEPA1*, *PTN*, *RAMP3*, *RBP1*, *RGCC*, *RHOC*, *SMAD1*, *SOX17*, *SOX4*, *SPARC*, *TCF4*, *UNC5B*, *VIM*. Stalk cell: *ACKR1*, *AQP1*, *C1QTNF9*, *CD36*, *CSRP2*, *EHD4*, *FBLN5*, *HSPB1*, *LIGP1*, *IL6ST*, *JAM2*, *LGALS3*, *LRG1*, *MEOX2*, *PLSCR2*, *SDPR*, *SELP*, *SPINT2*, *TGFBI*, *TGM2*, *TMEM176A*, *TMEM176B*, *TMEM252*, *TSPAN7*, *VEGFR1*, *VWF*.

We then used a gene set enrichment-based supervised classification method to calculate the tip-like score ($S_T$) and stalk-like score ($S_S$) of each CapEC (Extended Data Fig. 3e) using the R package AUCell[18]. Subsequently, we calculated the ratio of tip-like over stalk-like score as $R_{T/S} = (S_T + 1)/(S_S + 1)$ in pan-tumour CapEC subpopulations. CapECs with $R_{T/S} > 1$ were considered to be tip-like cells (V1, V7, V9, V12, V18 and V24), whereas those with $R_{T/S} < 1$ were considered to be stalk-like cells (V6, V16 and V23). Moreover, the $R_{T/S}$ of V0, V2 and V10 was close to 1. We therefore annotated them as transition cells between tip- and stalk-like cells.

### Cell trajectory inference

Diffusion Map[14], Slingshot[71], Monocle3[72] and VIA[73] were used to perform cell trajectory inference. Trajectory inference analyses based on different algorithms were used to corroborate each other. Diffusion Map is a spectral method for nonlinear dimensionality reduction and has been adapted to capture the differentiation trajectories. We calculated the diffusion components based on the CSS representation using the R package destiny[74]. Slingshot computes the pseudotime by proposing a simultaneous principal curves algorithm to construct smooth curves from the minimum spanning tree by the R package slingshot. Monocle3 generates the trajectory using the principal graph algorithm. VIA is a graph-based trajectory inference algorithm using lazy-teleporting random walks integrated with Markov chain Monte Carlo (MCMC) refinement, implemented by the Python package pyVIA. The predicted orders of Slingshot, VIA and Monocle3 were projected onto the UMAP space. CytoTRACE was used to assess the state of cell differentiation[15].

### RNA velocity

To restore the cell states of tumour angiogenesis, we first used velocyto.py to extract splicing information from the *.bam files generated by Cell Ranger. The resulting *.loom files were merged and transformed into h5ad format for further processing by Dynamo[75] and CellRank[76] with the default and recommended settings.

### HIF-1α activity evaluation

HIF-1α is the main TF responding to reduced oxygen availability[77,78]. We assessed the activity of HIF-1α based on three levels: (1) *HIF1A* transcription levels; (2) AUCell score of HIF-1α-target genes[79,80]; (3) DoRothEA activity of HIF-1α. DoRothEA[81] is a gene regulatory network containing signed TF–target gene interactions, and subsequently decoupleR was used to infer HIF-1α activity[82].

### TAPF identification and SOM

We first used SOM to identify overexpressed genes to characterize the functions of tip- and stalk-like cells. SOM is an unsupervised machine learning algorithm to translate the high-dimensional $N \times M$ expression data matrix ($N$, number of genes; $M$, number of cells) into a $K \times M$ metadata matrix ($K$, number of so-called metagenes) of reduced dimensionality[83]. We represented a 2D grid of size $K = 50 \times 50$ metagenes of which 8 overexpressed metagene signatures included 971 genes (Extended Data Fig. 4g). SOM was implemented by a customizable suite of single-cell R analysis tools (https://github.com/hloefflerwirth/scrat). Function enrichment analysis (Extended Data Fig. 4h) of overexpressed metagene signatures was performed using Metascape[84].

### PC activity analysis

We used a predefined quiescent-PC signature and angiogenic-PC signature[35] to evaluate PC activity by AUCell score[18] (Extended Data Fig. 7d). We then defined cells with scores higher than the upper quartile as the quiescent high ($Q^{high}$) group and cells with scores lower than the lower upper quartile as the quiescent low ($Q^{low}$) group. Differential expression analysis was performed between the two groups using the FindAllMarkers function (Fig. 4b).

### TradeSeq analysis

TradeSeq analysis[85] was implemented based on the LEC lineages inferred by Slingshot to identify the key TFs for LEC differentiation. The human TF list was obtained from AnimalTFDB[86]. First, we fitted the tradeSeq negative binomial generalized additivity model after estimating the optimal number of knots using the evaluate function. Next, the generalized additive model was fitted with the fitGAM function using nknots = 6. Subsequently, we compared the average expression of human TFs between two LEC lineages using the diffEndTest function. The $P$ value was corrected using the Benjamini–Hochberg adjustment ($q$ value).

### NicheNet analysis

We used NicheNet to perform stage-related communications among neovascular cells[87]. NicheNet uses the expression profile of specific sender cells and reviewer cells to construct the expressed LR interactions and screens ligands with high activities. We evaluated the communications between tip- and stalk-like cells at each angiogenic stage (SI, SII and SIII), respectively. We also applied a pseudo-bulk Spearman's correlation analysis to calculate the association within each LR pair to narrow the stage-specific LR pairs. Pseudo-bulk profiles were generated by calculating the sum of the transcript counts across all cells per sample. LR pairs with $P < 0.05$ were considered as stage-specific LR pairs (Supplementary Table 3).

### SCENIC analysis

We used SCENIC to identify the TFs driving the matPC trajectory[18]. SCENIC defines the regulon, which includes the TF and its potential target genes, for each cell population, and estimates the activity of the regulon. The activity level of the regulon indicates the predicted influence of the TF in the cell, with higher activity suggesting a greater influence. SCENIC analysis was run in the Python version pySCENIC. We used 'pyscenic grn', 'pyscience ctx' and 'pyscience aucell' to infer the gene regulatory network, regulon prediction and cellular enrichment (area under the curve, AUC) processed with the corresponding cells.

### SuperCell modelling

We sorted and integrated microenvironmental cells from the vasculature donors, including myeloid cells, T/NK cells, B cells and fibroblasts (Supplementary Fig. 5), as described above for extracting vascular cells. Next, we subgrouped the microenvironmental cells and annotated their subtypes on the basis of known gene markers. The SNN map was built based on the UMAP space, and the Louvain algorithm was used to cluster the cells. The myeloid subtypes included conventional type 1 dendritic cells (cDC1) (*BATF3* and *XCR1*), cDC2 (*CLEC10A* and *FCER1A*), DC3 (*LAMP3* and *CCR7*), tumour-associated neutrophils (*FCGR3B* and *CSF3R*), monocytes (*FCN1*), MDSCs (*S100A8* and *S100A9*), M2-like macrophages (*SEPP1* and *RNASE1*) and *SPP1*$^{+}$ TAMs (*SPP1*). T cells were first characterized by *CD4* and *CD8* expression as $CD4^{+}$, $CD8^{+}$ and double-positive, and subsequently classified as exhausted T cells (*PDCD1* and *CTLA4*), $T_{reg}$ cells (*FOXP3*), naive/memory T cells (*SELL* and *IL7R*), γδ T cells (*TRGC1*, *TRGC2* and *TRDC*), resident memory cells (*ZNF683* and *ITGAE*) and cytotoxic T cells (*GZMK*). NK cells were spatially close to T cells in the UMAP representation and were defined by *NKG7*. The subtypes of B cells included germinal centre (GC) B cells (*AICDA*, *MYBL1* and *SERPINA9*), naive B cells (*FCER2* and *TCL1A*), precursor memory B cells (*MYC* and *CD83*), memory B cells (*CD24* and *CD27*) and plasma cells (*MZB1*, *IGHA1*, *IGHJ1* and *JCHAIN*). Fibroblast subtypes comprised normal fibroblasts mainly from adjacent normal samples, inflammatory cancer-associated fibroblasts (*S100A10*, *S100A4* and *CXCL12*), adipocyte-like CAFs (*APOD*, *APOE* and *PTGDS*), myoid-like CAFs (*ACTA2* and *TAGLN*) and matCAFs (*PLAU*, *MMP1* and *MMP3*). We next used SuperCell algorithms to fit metacell models for microenvironmental and vascular cells[88]. SuperCell is based on a computational coarse-graining framework that can group highly similar cells into metacells. In brief, the $k$-nearest neighbours graph of scRNA-seq data was first constructed, and metacells were built by merging single cells with very high internal connectivity. Finally, a metacell gene expression profile was computed by averaging gene expression within metacells. The graining level ($\gamma$) was set to 10.

### Cell–cell interaction analysis

We used CellPhoneDB[89] to infer unbiased cell–cell interactions between (1) MCs and VEC/LEC and (2) microenvironmental cells (myeloid, T, NK

and B cells and fibroblasts) and vasculature cells, respectively. The metacell expression profiles were used as the input of CellPhoneDB. CellPhoneDB used a statistical framework to infer cell-type-enriched LR interactions between different cell types in scRNA-seq data. The potential interaction strength between the two cell subsets was estimated based on the expression of their respective LR pairs. The enriched LR interactions were calculated using permutation tests. We retained those LR pairs with expression percentage > 25% in each cell type.

### Cell type deconvolution of bulk RNA-seq

A cell type deconvolution of The Cancer Genome Atlas (TCGA) data was performed using the BayesPrism[90]. Feature genes for cell types of VECs, LECs and MCs were calculated using Seurat FindAllMarkers. Genes with average $\log_2$-transformed fold change of >1 and the percentage of cells within the cluster that expressed each gene of >0.25 were used in the subsequential analysis (Supplementary Table 2). Mitochondrial and ribosomal protein genes were removed from the feature gene list. For each of the cell types, mean raw counts were calculated to generate a gene expression profile for a single-cell reference. The single-cell reference was used as input alongside the TCGA bulk data into a Bayesian deconvolution method to estimate the cell fractions of bulk data (Supplementary Table 4).

### Cell type deconvolution of spatial transcriptome data

We applied conditional autoregressive deconvolution (CARD)[42] to the in-house and external spatial data produced using the 10x Genomics Visium platform (Supplementary Table 1). Input data included spatial transcriptomics count data, along with spatial location information, and scRNA-seq count data, along with meta information indicating all the cell types and the sample information for each cell. We assessed the expression intensity of an LR pair by calculating the geometric mean between ligand and receptor.

### Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

## Data availability

Raw sequencing data, processed scRNA and spatial transcriptome datasets reported in this paper have been deposited at the Genome Sequence Archive at the National Genomics Data Center (Beijing, China) under BioProject PRJCA018695. The data deposited and made public are compliant with the regulations of the Ministry of Science and Technology of the People's Republic of China. Integrated gene expression data can be accessed through an online data browser (http://resource.yin-lab.com/Panvascular/). Public scRNA datasets and spacial transcriptomiocs data used in this study are shown in Supplementary Table 1 and TCGA datasets are from https://portal.gdc.cancer.gov/.

## Code availability

Custom codes have been deposited at GitHub (https://github.com/bio-Pixel/panVC) and Zenodo[91] (https://zenodo.org/records/11188740).


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**Author contributions** M.Y. and X.P. conceived the presented idea. X.P. and X.L. designed and performed bioinformatic analyses. L.D. and M.Y. performed the mIHC staining and survival analysis. H. Shao and Y.W. constructed the animal model. M.Y., M.Z. and H. Sun collected tissue samples. T.L., L.Z., X.Z., L.H., W.S., Z.F., J.S. and Y.H. collected single-cell data. X.P., X.L., L.D. and M.Y. wrote the manuscript. M.Y. supervised this project. All of the authors discussed the results and contributed to the final manuscript.

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**Additional information**

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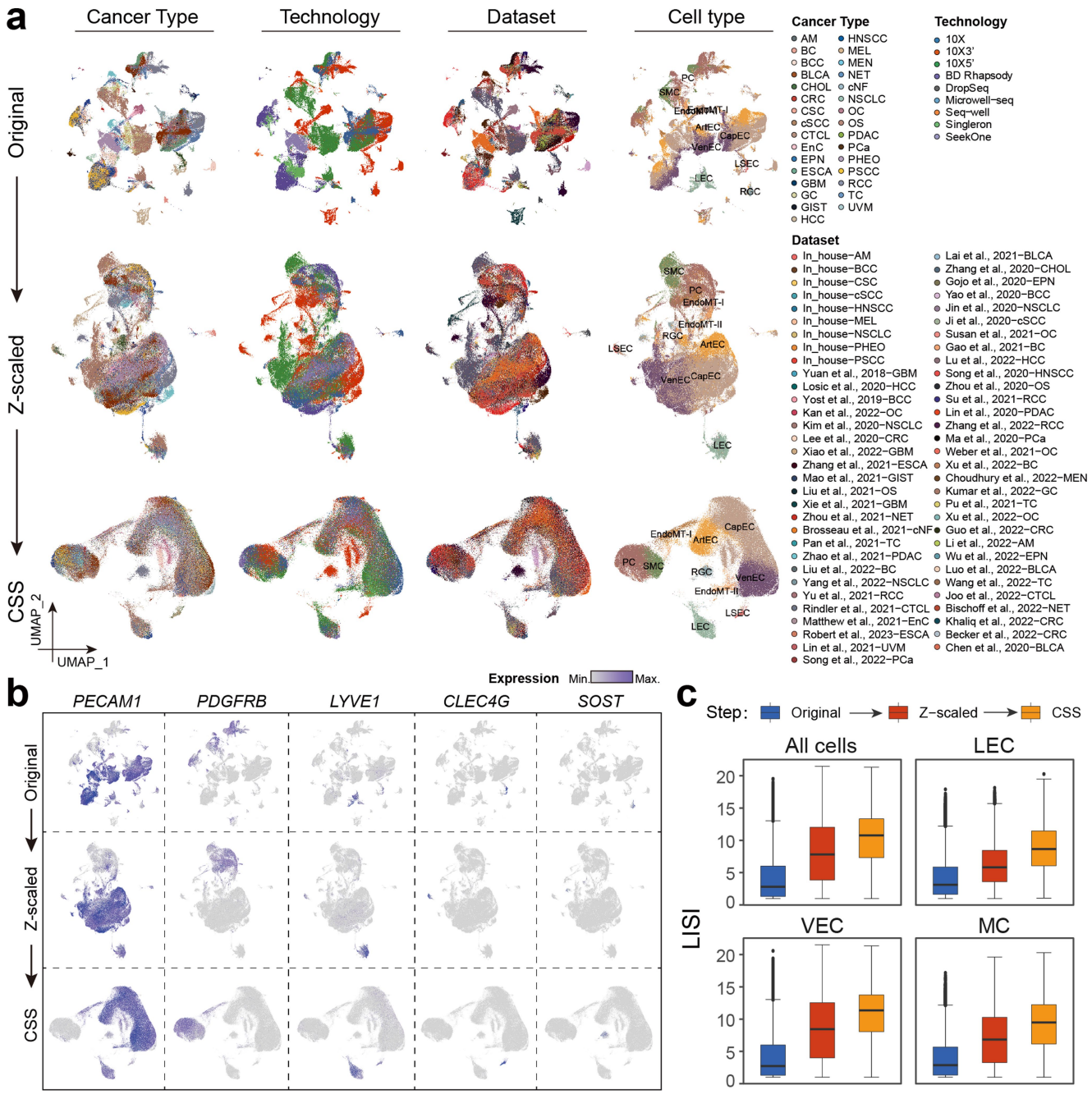
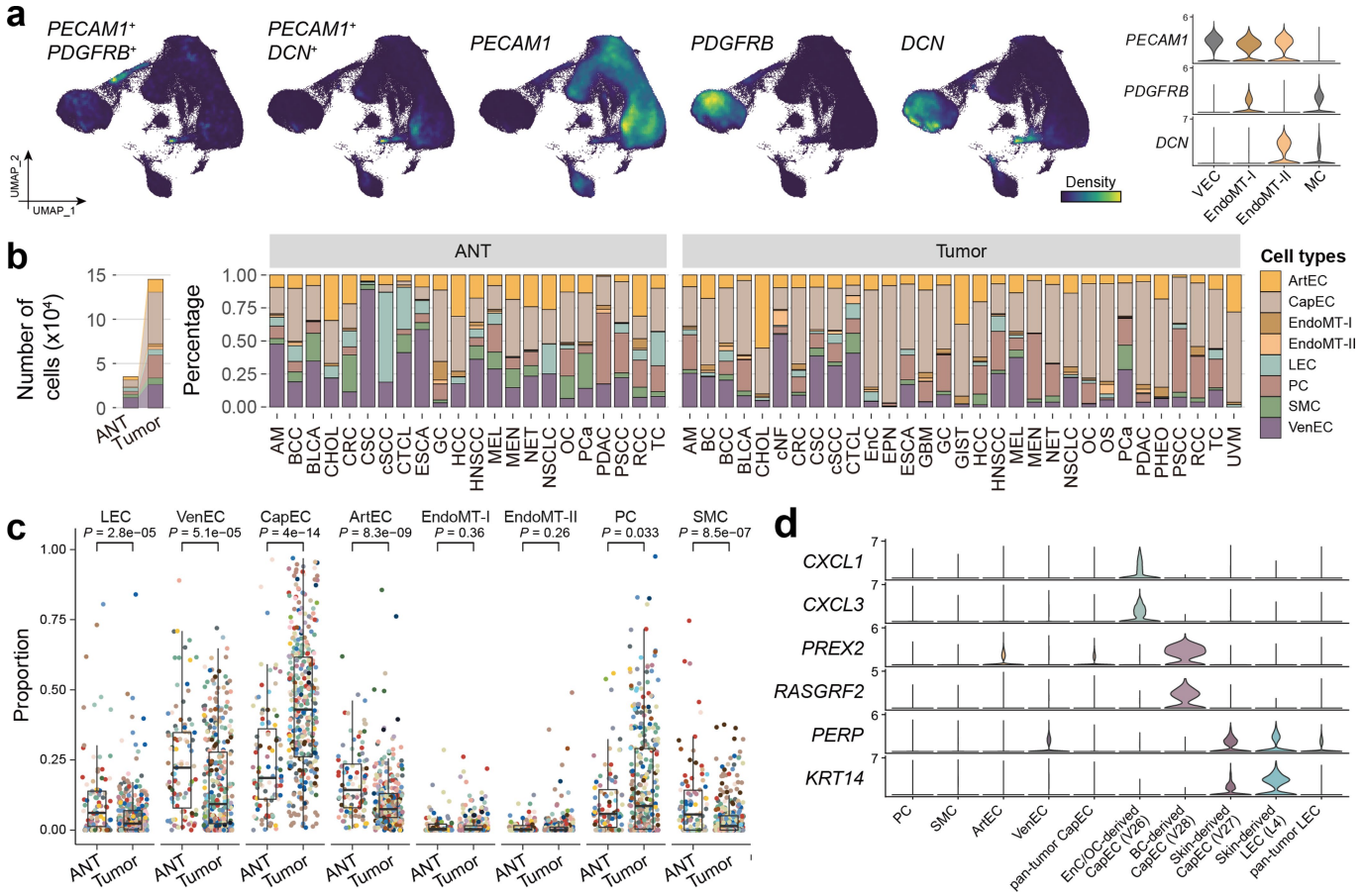
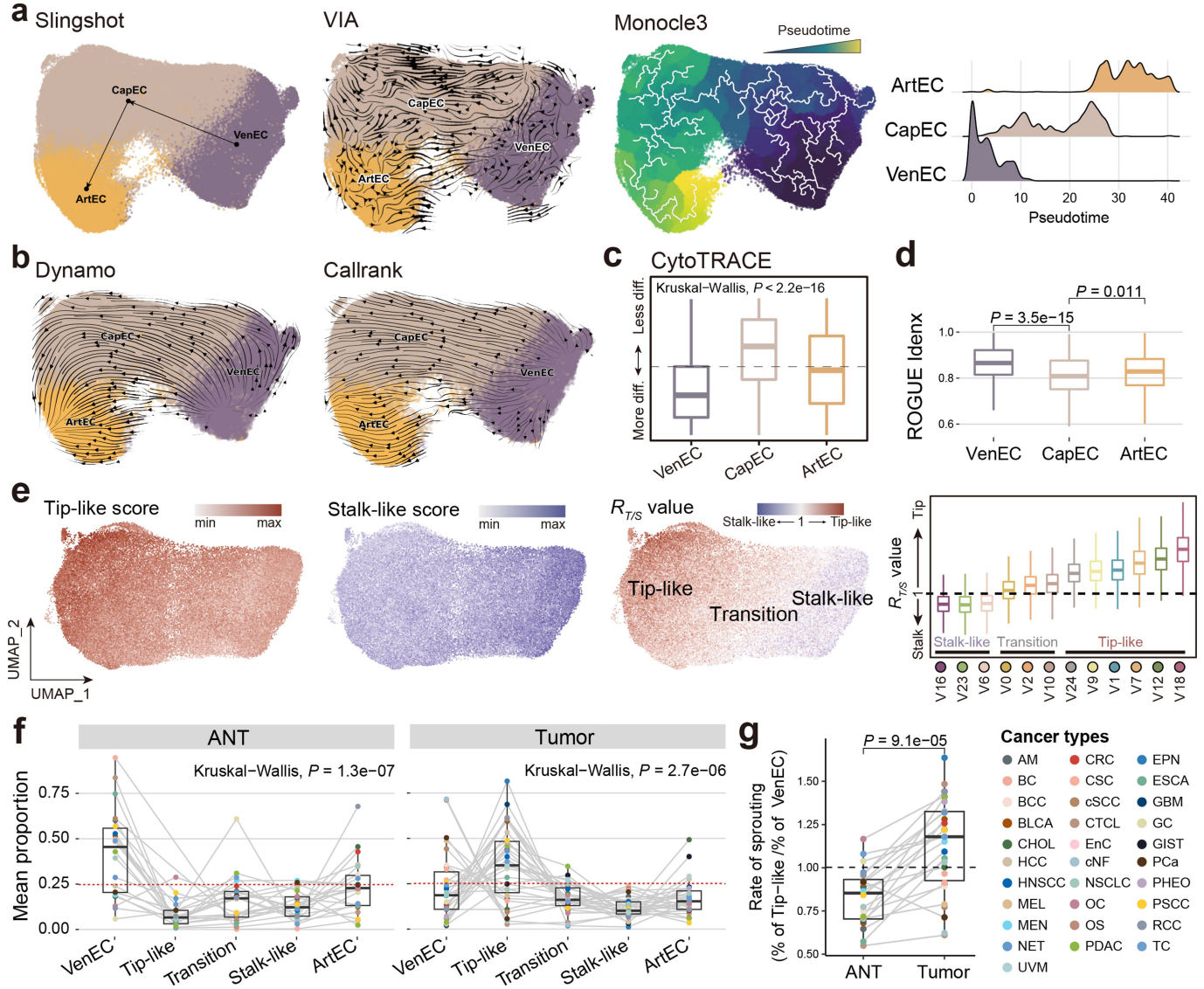


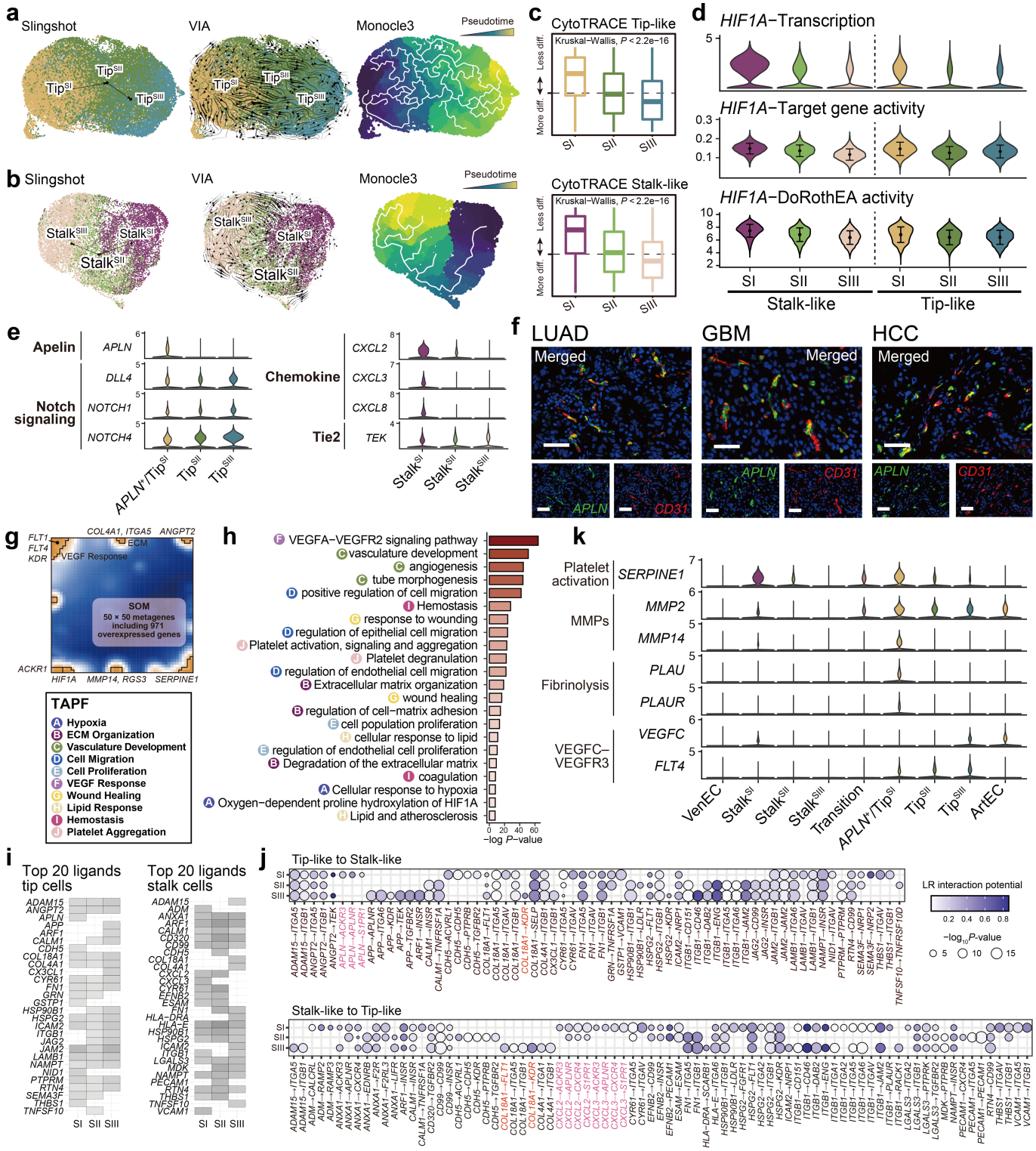

**Extended Data Fig. 1 | Data integration of vascular cells. a**. UMAP plots showing the distribution of vascular cells following the integration steps (Original count to Z-scaled to CSS). After CSS processing, cells produced by different platforms were clustered obviously. **b**. UMAP plot showing selected marker gene expression of cell types. **c**. Boxplot showing the LISI (Local Inverse Simpson's Index) values after each integration step (n = 116,958 cells). For boxplots, centre line represents median, box limits indicate upper and lower quartiles, and whiskers extend 1.5 times the interquartile range.




**Extended Data Fig. 2 | Data integration of vascular cells. a**. UMAP plots showing the joint density estimation of *PECAM1-PDGFRB* and *PECAM1-DCN*. Violin plot showing the expression levels of *PECAM1*, *PDGFRB*, and *DCN* across VECs, EndoMT cells, and MCs. **b**. Bar plot showing the total cell number (right) and percentages (left) of vascular cells in ANT and tumour tissues. **c**. Box plot showing the proportion of vascular cells in each sample. Dots represent single-cell samples (ANT, n = 99, Tumour, n = 338), coloured by cancer types. *P*-value was calculated by two-sided Mann–Whitney U test. **d**. Violin plot showing the expression levels of selected markers in VEC/LEC subpopulations with similar tissue origin. For boxplots, centre line represents median, box limits indicate upper and lower quartiles, and whiskers extend 1.5 times the interquartile range.

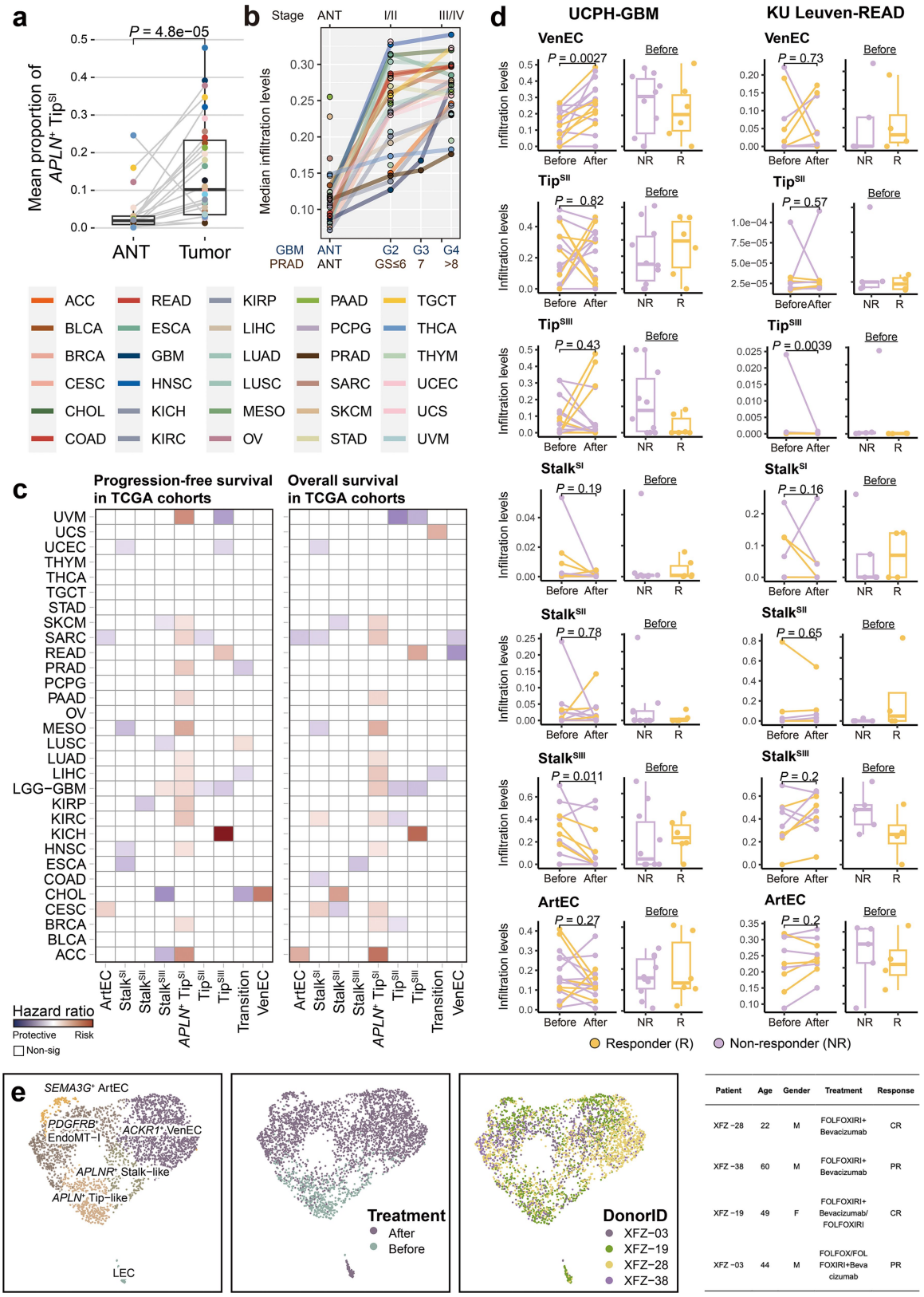



**Extended Data Fig. 3 | Tumour angiogenesis initiating from VenEC.** **a**. Visualization of angiogenesis trajectories inferred by Slingshot, VIA, and Monocle3. **b**. Velocity-based cell state inference supports VenEC as angiogenesis initiation. **c**. Boxplot showing the CytoTRACE scores among VenEC (n = 37,510), CapEC (n = 61,400), and ArtEC (n = 18,048). Less diff., less differentiated; More diff., more differentiated. **d**. Boxplot showing cell purity for VenEC (n = 37,510), CapEC (n = 61,400), and ArtEC (n = 18,048) by ROGUE. **e**. UMAP plot showing the tip-like, stalk-like, and $R_{T/S}$ scores across CapECs. Boxplot showing the distribution of $R_{T/S}$ values among CapEC subclusters. Six subclusters (V1 [n = 6,351], V7 [n = 5,376], V9 [n = 5,268], V12 [n = 4,831], V18 [n = 3,687], and V24 [n = 3,291]) were annotated as "tip-like" CapECs, three (V6 [n = 5,517], V16 [n = 4,399], and V23 [n = 3,361]) as "stalk-like" CapECs, and three (V0 [n = 8,194], V10 [n = 4,946], and V2 [n = 6,179]) in a "Transition" state between tip- and stalk-like cells. **f**. Box plot showing the mean proportions of each cancer type across VEC subtypes in ANT and tumour tissues, respectively. Dots represent cancer types (n = 31). Briefly, we first calculated the proportion of each cell type in each sample and then calculated the average proportion in each cancer. **g**. Boxplot showing the sprouting rate between ANT and tumour tissues. Dots represent cancer types (n = 31). Sprouting rate was expressed as the ratio of tip-like cell to VenEC. For boxplots, centre line represents median, box limits indicate upper and lower quartiles, and whiskers extend 1.5 times the interquartile range. Statistical analysis was performed using two-sided Mann–Whitney U test (**d** and **g**) and Kruskal-Wallis test (**c** and **f**).




**Extended Data Fig. 4** | See next page for caption.

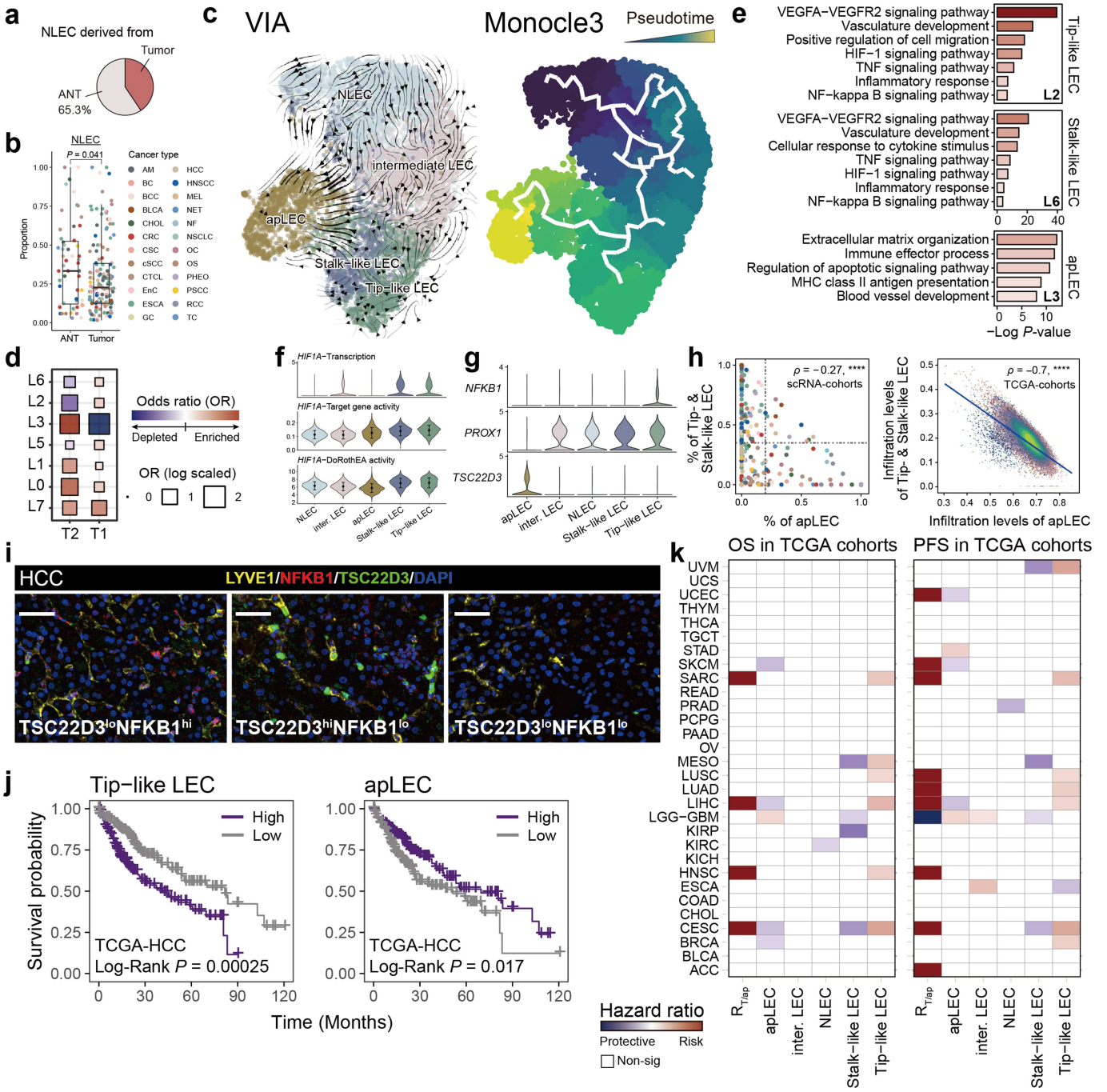
**Extended Data Fig. 4 | Tip-like and stalk-like cells in TVM. a**-**b**. Visualization of trajectories inferred by Slingshot, VIA, and Monocle3 in tip-like (**a**) and stalk-like (**b**) cells, respectively. **c**. Boxplot showing the CytoTRACE scores among differentiation stages of tip-like (top) and stalk-like (bottom) cells (tip cells: SI, n = 9,063, SII, n = 13,390, SIII, n = 6,351; stalk cells: SI, n = 4,399, SII, n = 3,361, SIII, n = 5,517). **d**. Violin plots showing the HIF1A activity at transcription levels (top), AUCell score of target genes (middle), and DoRothEA activity (bottom). Error bar represents mean ± standard deviation. **e**. Violin plots showing the expression of staged markers. **f**. Representative mIHC of *APLN*$^{+}$ Tip$^{SI}$ markers in LUAD from n = 10 independent biological replicates, GBM from n = 11 independent biological replicates, and HCC from n = 10 independent biological replicates. Scale bar, 50 μm. **g**. SOM (50 × 50 grid) constructed from single-cell transcriptomes of tip-like and stalk-like cells showing scaled metagene expression. Black lines demarcate eight overexpressed metagene-signatures comprising 971 overexpressed genes. **h**. Function enrichment of overexpressed metagenes interrogated by SOM, resulting in ten TAPFs. *P*-values are calculated based on the cumulative hypergeometric distribution. **i**-**j**. Tip-to-stalk communications in angiogenic stages. Heatmap showing the activities of top 20 ligands (**i**). Dot plot showing the stage-related LR pairs (**j**), sized by the pseudobulk Spearman's correlation *P*-value and coloured by the interaction potential predicted using NicheNet. **k**. Violin plot showing the expression levels of genes associated with TAPFs. For boxplots, centre line represents median, box limits indicate upper and lower quartiles, and whiskers extend 1.5 times the interquartile range. Statistical analysis was performed using Kruskal-Wallis test (**c**).



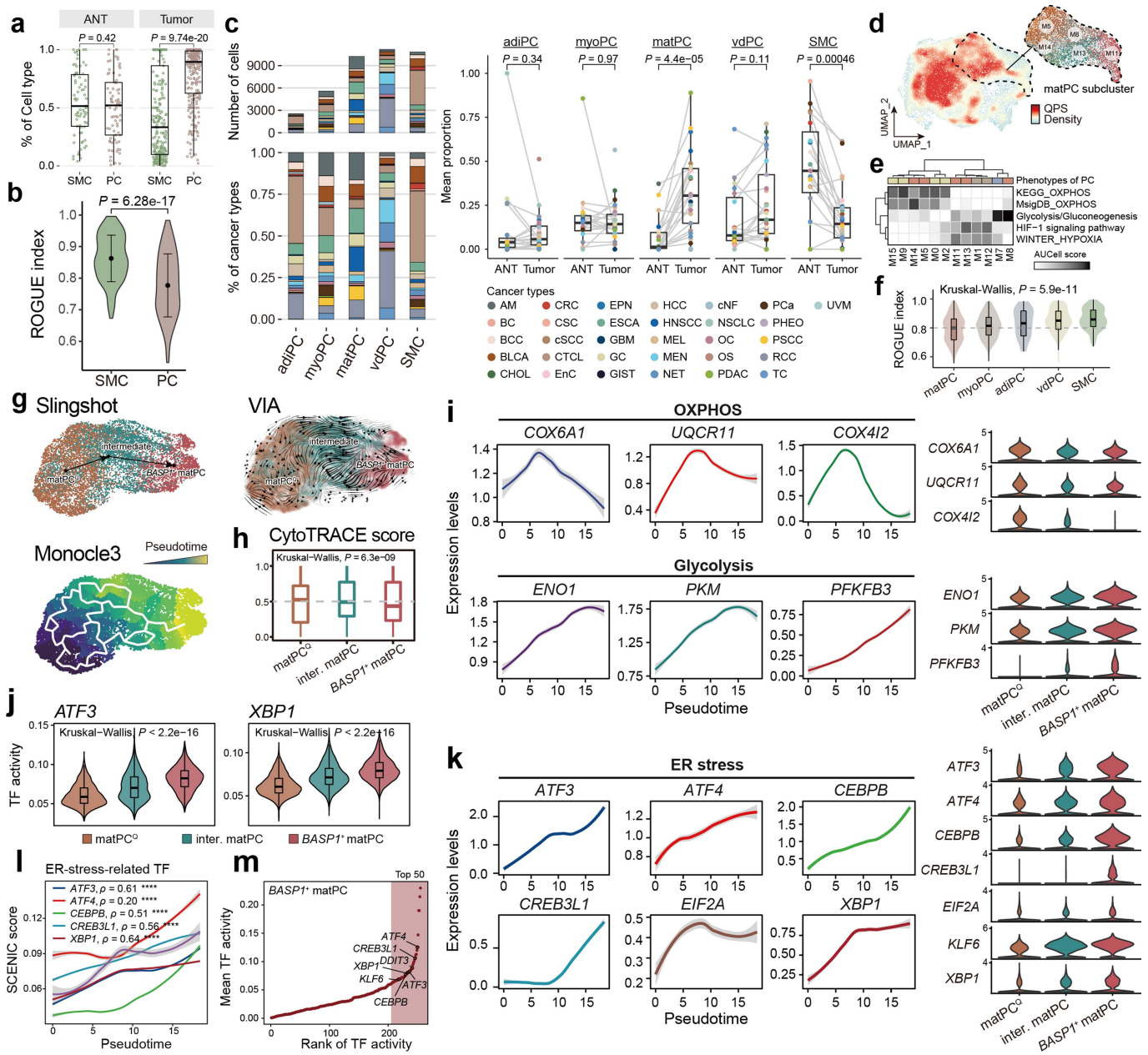

| Patient | Age | Gender | Treatment | Response |
|---|---|---|---|---|
| XFZ−28 | 22 | M | FOLFOXIRI+ Bevacizumab | CR |
| XFZ−38 | 60 | M | FOLFOXIRI+ Bevacizumab | PR |
| XFZ−19 | 49 | F | FOLFOXIRI+ Bevacizumab/ FOLFOXIRI | CR |
| XFZ−03 | 44 | M | FOLFOX/FOL FOXIRI+Beva cizumab | PR |

**Extended Data Fig. 5** | See next page for caption.

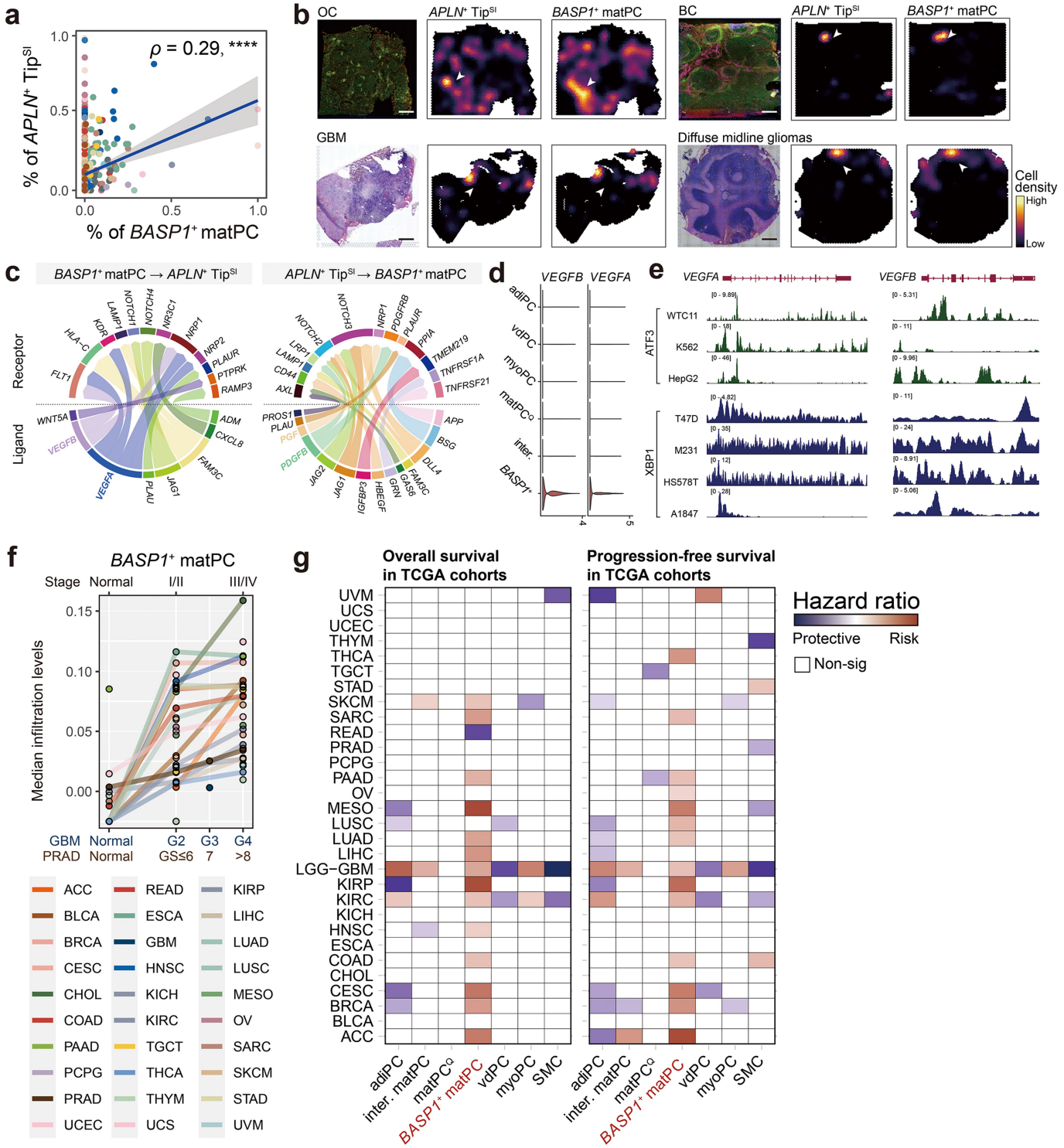
**Extended Data Fig. 5 | Clinical implications of VECs. a**. Box plot showing the mean proportions of *APLN*$^{+}$ Tip$^{SI}$ in each cancer type between ANT and tumour tissues. Dots represent cancer types (n = 31). **b**. The median infiltration levels of *APLN*$^{+}$ Tip$^{SI}$ in each TCGA cancer type across ANT, stage I/II, and stage III/IV samples. GBM was graded by G2, G3, and G4. PRAD was graded by Gleason score (GS). We performed the two-sided Mann–Whitney U test on the mean infiltration levels of *APLN*$^{+}$ Tip$^{SI}$ between stage I/II and ANT samples in each cancer type. Different stages of the same cancer are connected using coloured lines when $P < 0.05$. **c**. Heatmap showing the effect of VEC infiltration on OS (right) and PFS (left). **d**. Variations of VEC infiltration levels following bevacizumab treatment. Boxplots showing differences between pre-treatment responders (R) and non-responders (NR) in UCPH-GBM (N, n = 6, NR, n = 10), KU Leuven-READ (N, n = 6, NR, n = 5), and SYSU-READ (n = 4). **e**. UMAP plot showing the cell types (left), treatment time points (middle), and patients (right) in SYSU-READ. Table showing the patients' characteristics. For boxplots, centre line represents median, box limits indicate upper and lower quartiles, and whiskers extend 1.5 times the interquartile range. Statistical analysis was performed using two-sided Mann–Whitney U test (**a** and **d**).



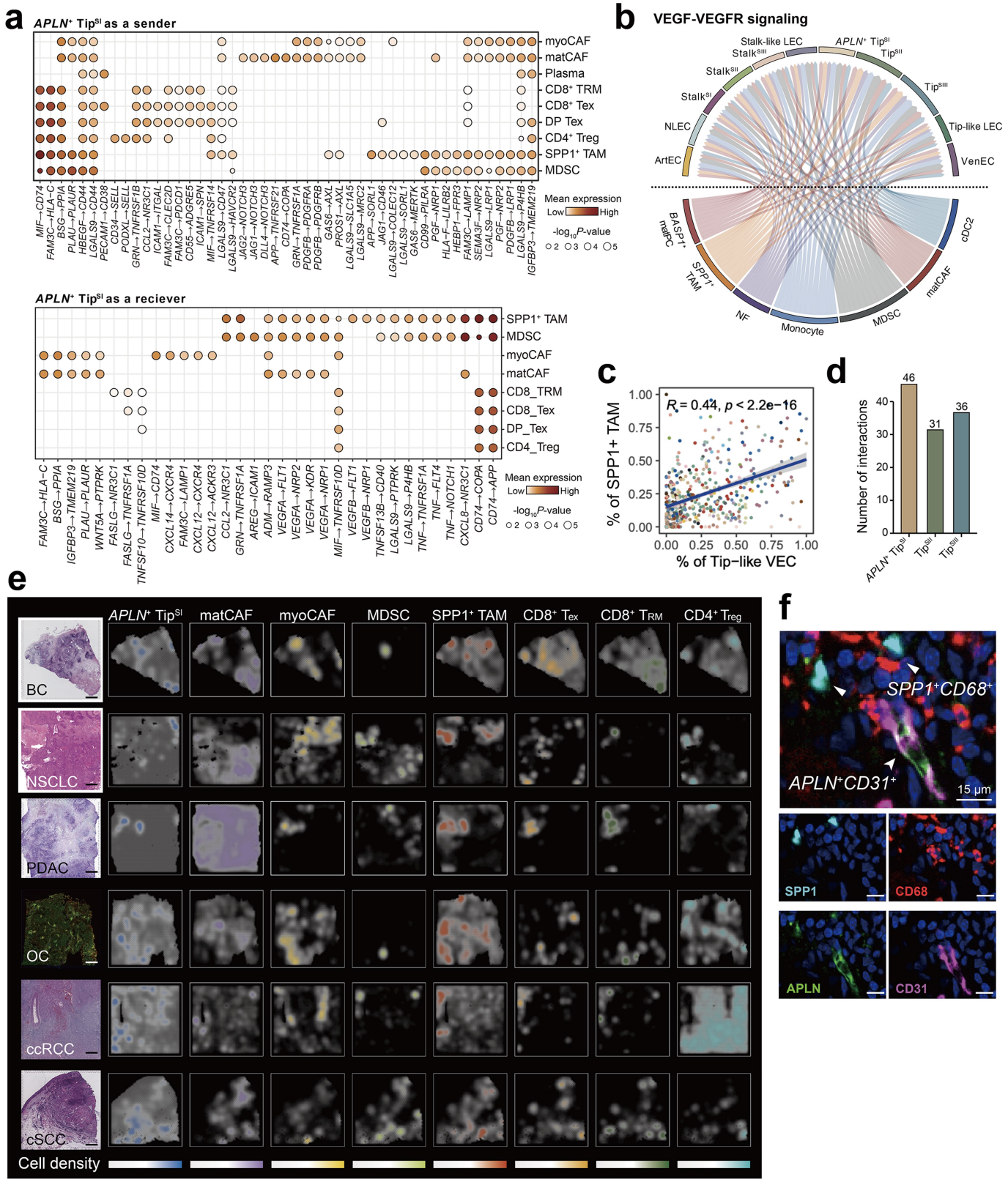

**Extended Data Fig. 6 | LEC in TVM. a**. Pie chart showing the percentage of NLEC detected from ANT and tumour tissues. **b**. Box plot showing the proportion of NLEC in each sample. Dots represent single-cell samples (ANT, n = 83, Tumour, n = 253), coloured by cancer types. **c**. Visualization of the LEC trajectories inferred by VIA and Monocle3. **d**. Bubble plot showing the odds ratio of LEC subpopulations enriched in LEC lineages. Odds ratios are computed using Fisher's exact test. **e**. Function enrichment of marker genes of apLEC, tip-like LEC, and stalk-like LEC. *P*-values are calculated based on the cumulative hypergeometric distribution. **f**. The activity of *HIF1A* across LEC subtypes (NLEC, n = 2,872; inter. LEC, n = 2,574; apLEC, n = 1,309; Stalk-like LEC, n = 975; Tip-like LEC, n = 1553). Error bar represents mean ± standard deviation. **g**. Violin plots showing the expression of *PROX1*, *NFKB1*, and *TSC22D3* among LECs. **h**. Correlation of the infiltration levels of apLEC and neolymphatic vessels in the scRNA-seq dataset (left) and TCGA bulk-seq dataset (right). Pearson's correlation coefficient and its *P*-value are shown. ****, $P < 2.2\times10^{-16}$. **i**. Representative examples of mIHC staining with anti-LYVE1 (yellow), anti-NFKB1 (red), and anti-TSC22D3 (green) in HCC from n = 12 independent biological replicates. Scale bar, 50 μm. **j**. Kaplan–Meier curves of OS for TCGA-HCC samples. Tumour samples were stratified by the median infiltration levels of tip-like LECs (left) and apLECs (right), respectively. **k**. Heatmap showing the effect of LEC infiltration and $R_{T/ap}$ on OS (left) and PFS (right). For boxplots, centre line represents median, box limits indicate upper and lower quartiles, and whiskers extend 1.5 times the interquartile range. Statistical analysis was performed using two-sided Mann–Whitney U test (**b**) and log-rank test (**j**).



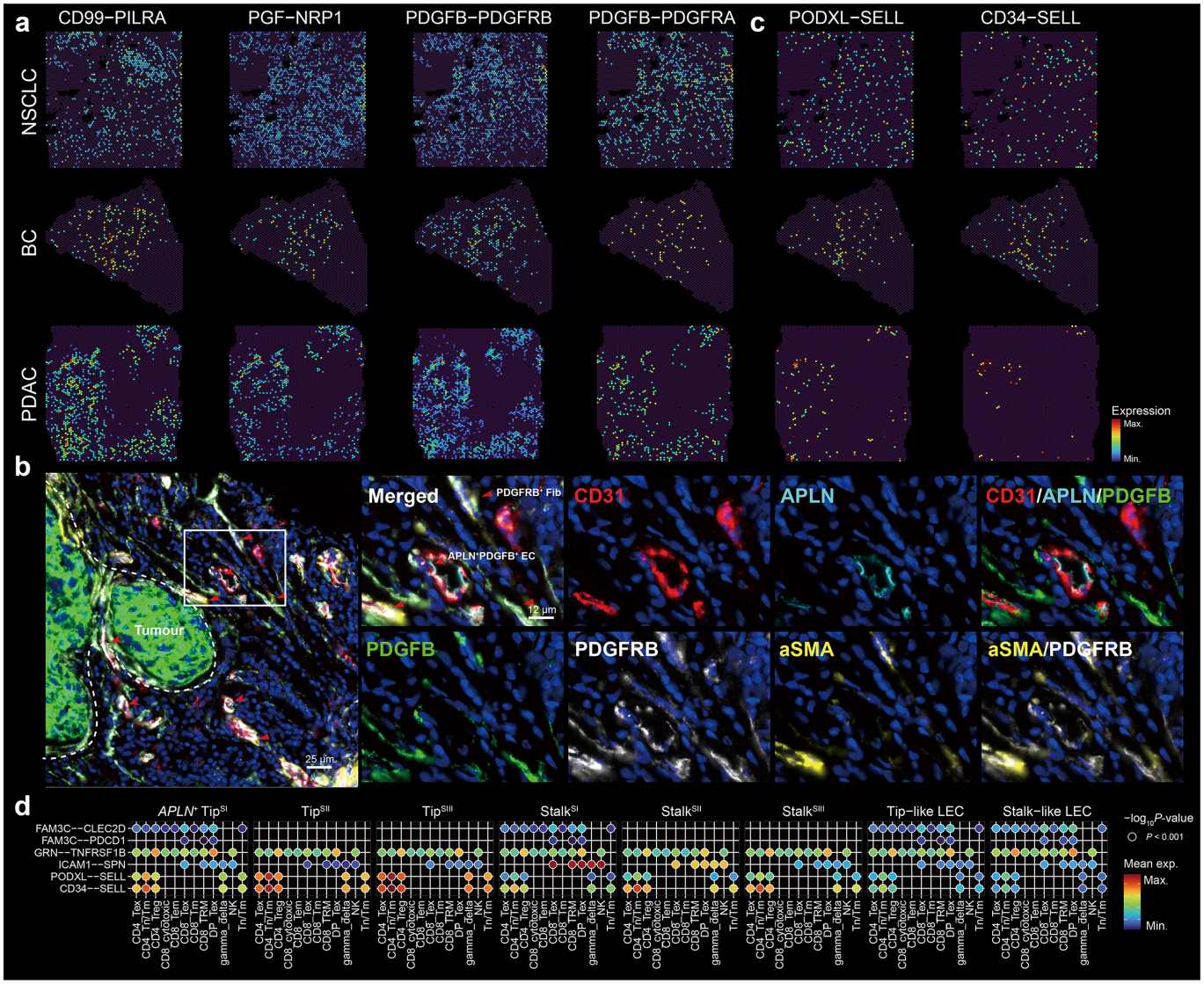

**Extended Data Fig. 7 | Characteristics of PC phenotypes. a**. Boxplot showing the proportions of SMCs and PCs in ANT (n = 77) and tumour (n = 265) samples, respectively. **b**. Violin plot showing cell purity for SMC (n = 10,819) and PC (n = 29,572) by ROGUE. Error bar represents mean ± standard deviation. **c**. Bar plot showing the cell number (top) and percentages (bottom) of cancer types across MC subtypes. Box plot showing the mean proportions of each cancer type between ANT (n = 77) and tumour (n = 265) samples across MC subtypes. **d**. UMAP plot showing the density of quiescent score. MatPCs showing heterogeneous quiescent states. **e**. Heatmap showing the activities of energy metabolism pathways among PC subclusters. **f**. Boxplot showing cell purity of MC subtypes by ROGUE. SMC, n = 10,819, adiPC, n = 2,562, matPC, n = 10,230, myoPC, n = 5,599, vdPC, n = 11,181. **g**. Visualization of the matPC trajectories inferred by Slingshot, VIA, and Monocle3. **h**. Boxplot showing the CytoTRACE score among matPC subtypes. *BASP1*$^{+}$ matPC, n = 2,060, inter. matPC, n = 4,211, matPC$^{Q}$, n = 3,959. **i**. Loess regression-smoothened gene expression of the oxidative-stress genes (y-axis) associated with in pseudotime. Violin plot showing the expression levels of corresponding genes. **j**. Violin plot showing the SCENIC scores of *ATF3* (right) and *XBP1* (left). *BASP1*$^{+}$ matPC, n = 2,060, inter. matPC, n = 4,211, matPC$^{Q}$, n = 3,959. **k-l**. Loess regression-smoothened expression (y-axis in **k**) and SCENIC activity (y-axis in **f**) in pseudotime. Spearman's correlation coefficient and its *P*-value are shown in **f**. ****, $P < 0.0001$. **m**. TF activity in *BASP1*$^{+}$ matPC. ER-stress-related TFs were labelled. Shaded region represents the 95% confidence intervals (**i**, **k**, and **l**). For boxplots, centre line represents median, box limits indicate upper and lower quartiles, and whiskers extend 1.5 times the interquartile range. Statistical analysis was performed using two-sided Mann–Whitney U test (**a**, **b**, and **c**) and Kruskal–Wallis test (**f**, **h**, and **j**).




**Extended Data Fig. 8 | *BASP1*$^+$ matPC is involved in tumour angiogenesis.** **a**. Correlation between the proportion of *BASP1*$^+$ matPC and *APLN*$^+$ Tip$^{SI}$. Shaded region represents the 95% confidence intervals. Spearman's correlation coefficient and its *P*-value are shown. Dots coloured by cancer types. **b**. Co-localization between *APLN*$^+$ Tip$^{SI}$ and *BASP1*$^+$ matPC at spatial positions. Scale bar, 1 mm. **c**. LR pairs between *APLN*$^+$/Tip$^{SI}$ and *BASP1*$^+$ matPC. **d**. Violin plot showing the expression levels of VEGF across PC phenotypes. **e**. Genome browser view showing the binding of XBP1 and ATF3 in the genome region of *VEGFA/VEGFB*. ChIP-seq data were obtained from public datasets (XBP1: GSE49952 [BC cell lines: T47D, M231, and HS578T] and GSE157117 [OC cell line: A1847]; ATF3: ENCFF020THR [HCC cell line: HepG2], ENCFF341RJA [human induced pluripotent stem cell: WTC11], and ENCFF561SGX [leukaemia cell line: K562]). **f**. The median infiltration levels of *BASP1*$^+$ matPC in each TCGA cancer type across ANT, stage I/II, and stage III/IV samples. GBM was graded by G2, G3, and G4. PRAD was graded by Gleason score (GS). We performed the two-sided Mann–Whitney U test on the mean infiltration levels of *BASP1*$^+$ matPC between stage I/II and ANT samples in each cancer type. Different stages of the same cancer are connected using coloured lines when $P < 0.05$. **g**. Heatmap showing the effects of MC infiltration on OS (right) and PFS (left).




**Extended Data Fig. 9 | Cell interactions during tumour angiogenesis. a.** Dot plot showing the LR pairs between TME cells and *APLN*$^{+}$ Tip$^{SI}$ when *APLN*$^{+}$ Tip$^{SI}$ is a sender (top)/receiver (bottom). **b.** Circos plot showing the VEGF-VEGFR interactions between vascular cells and TME cells. **c.** Correlation between the proportion of *SPP1*$^{+}$ TAM and tip-like VECs. Dots coloured by cancer types. Shaded region represents the 95% confidence intervals. Spearman's correlation coefficient and its *P*-value are shown. **d.** Bar plow showing the number of interactions between *SPP1*$^{+}$ TAM and tip cells. **e.** Co-localization among *APLN*$^{+}$ Tip$^{SI}$, *SPP1*$^{+}$ TAM, MDSC, myoCAF, matCAF, exhausted T cells, and $T_{reg}$ at spatial positions from ST data of BC (n = 1 case), NSCLC (n = 1 case), PDAC (n = 1 case), OC (n = 1 case), ccRCC (n = 1 case), and cSCC (n = 1 case). Scale bar, 1 mm. **f.** Representative mIHC of *APLN*$^{+}$ tip cells and *SPP1*$^{+}$ TAM in NSCLC from n = 7 independent biological replicates. Arrows depict the specific cell types.




**Extended Data Fig. 10 | LR pairs associated with *APLN*$^{+}$ Tip$^{SI}$. a**. Spatial plot showing the expression intensity of LR pairs of PAC interactions. *APLN*$^{+}$ Tip$^{SI}$ can recruit PACs via CD99-PILRA, PGF-/PDGFB-related LR pairs. **b**. Representative mIHC of *PDGFB*-expressing *APLN*$^{+}$ tip cells and *PDGFRB*-expressing fibroblasts in CSC from n = 8 independent biological replicates. Arrows depict the specific cell types. **c**. Spatial plot showing the expression intensity of LR pairs of Treg homing *PODXL/CD34-SELL*. **d**. Neovascularization can interact with T cells via GRN-TNFRSF1B, CD34/PODXL-SELL, ICAM1-SPN, and FAM3C-PDCD1/CLEC2D pairs. Expression intensity was calculated by the geometric mean of an LR pair (**a** and **c**).

nature portfolio

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Last updated by author(s): May 19, 2024

# Reporting Summary

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## Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

| n/a | Confirmed |
|---|---|
| ☐ | ☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement |
| ☐ | ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly |
| ☐ | ☒ The statistical test(s) used AND whether they are one- or two-sided<br>*Only common tests should be described solely by name; describe more complex techniques in the Methods section.* |
| ☒ | ☐ A description of all covariates tested |
| ☐ | ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons |
| ☐ | ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals) |
| ☐ | ☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted<br>*Give P values as exact values whenever suitable.* |
| ☒ | ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings |
| ☒ | ☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes |
| ☐ | ☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated |

*Our web collection on statistics for biologists contains articles on many of the points above.*

## Software and code

Policy information about availability of computer code

| | |
|---|---|
| Data collection | SCOPE-tools (version 0.18), SeekOneTools (version 1.2.1), CellRanger (version 6.1.2), BD Rhapsody™ Sequence Analysis Pipeline (version1.9.1) and SpaceRanger (version 2.1.1) were used for data collection. |
| Data analysis | All bioinformatics analyses are based on publicly available software. R (version 4.1.3) and Python (version 3.8.13) were used for data analysis. Single-cell RNA-seq data was analyzed using Seurat (version 4.2.1), Simspec (version 0.0.0.9000), Harmony (version 0.1.0), scrat (version 2.1.0), nichenetr (version 1.1.1), destiny (version 3.8.1), slingshot (version 2.2.1), Monocle 3 (version 1.2.9), CytoTRACE (beta version), pyVIA (version 0.1.88), ggplot2 (version 3.3.6), ggthemes (version 4.2.4), pheatmap (version 1.0.12), ROGUE (version 1.0), BayesPrism (version 2.0), tradeSeq (version 1.8.0), pySCENIC (version 0.12.1), CARD (version 1.1), and cellphonedb (version 2.1.7). No new algorithms were developed for this manuscript. Custom codes have been deposited in GitHub (https://github.com/bio-Pixel/panVC) and Zenodo (https://zenodo.org/records/11188740). |

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

## Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Raw sequencing data, processed scRNA and spatial transcriptome datasets reported in this paper have been deposited at the Genome Sequence Archive at the National Genomics Data Center (Beijing, China) under the BioProject ID PRJCA018695 (https://ngdc.cncb.ac.cn/bioproject/browse/PRJCA018695). Public scRNA-seq datasets were reanalyzed, including GSE103224, GSE112271, GSE123813, GSE130000, GSE131907, GSE132465, GSE135045, GSE135337, GSE138709, GSE141460, GSE141526, GSE142784, GSE143423, GSE144236, GSE144735, GSE147082, GSE148673, GSE149614, GSE150321, GSE152048, GSE152938, GSE154778, GSE156632, GSE157703, GSE158937, GSE160269, GSE162115, GSE162454, GSE162631, GSE162708, GSE163028, GSE163203, GSE165399, GSE167036, GSE171145, GSE171306, GSE173205, GSE173682, GSE173950, GSE176029, GSE176031, GSE180286, GSE183655, GSE183904, GSE184362, GSE184880, GSE188711, GSE189889, GSE189939, GSE190888, GSE191288, GSE195674, GSE196303, GSE198745, GSE200997, GSE201348, and PRJNA662018. Public spatial transcriptome data were from Human Tumor Atlas Network (accession number: HT306P1, https://humantumoratlas.org/explore), 10x genomics datasets (https://support.10xgenomics.com/spatial-gene-expression/datasets/1.2.0/V1_Human_Invasive_Ductal_Carcinoma and https://support.10xgenomics.com/spatial-gene-expression/datasets/1.2.0/Parent_Visium_Human_OvarianCancer), and GEO (accession number: GSM4565826, GSM5924039, GSM6177599, GSM5833528, and GSM5833536, https://www.ncbi.nlm.nih.gov/geo/). The Cancer Genome Atlas Program (TCGA) data was from https://portal.gdc.cancer.gov/.

## Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

| | |
|---|---|
| Reporting on sex and gender | The findings from our apply to both sexes. The Gender information for our own sequencing data was determined based on self-reporting. Gender was not considered in study design. |
| Reporting on race, ethnicity, or other socially relevant groupings | The samples for our own sequencing data are all of Asian origin. It's essential to clarify that our study's primary focus was not on comparing tumor microenvironments across diverse racial, ethnic, or other socially relevant groups. Consequently, we did not collect information related to these specific factors during the research. |
| Population characteristics | Patients pathologically diagnosed with nine types of cancer were enrolled in this study, including acral melanoma (n = 14, age[mean ± sd]: 67.85±10.66), basal cell carcinoma (n = 8, age: 67.5±5.96), cutaneous squamous cell carcinoma (n = 2, age: 66±1.41), melanoma (n = 9, age: 45.00±21.08), cervical squamous cell carcinoma (n = 2, age: 50.50±4.95), non-small cell lung cancer (n = 7, age: 55.71±9.48), head and neck squamous cell carcinoma (n = 14, age: 52.57±9.21), pheochromocytoma (n = 6, age: 36.83±13.26), and penile squamous cell carcinoma (n = 4, age: 59.25±11.03). Detailed clinical information can be found in the Supplementary Table 1B. |
| Recruitment | Patients meeting the pathological diagnosis of the interested cancer type and agreed to sign the consent file were recruited in our study without any self-selection. All samples were obtained from Chongqing University Three Gorges Hospital, Central South University Xiangya Hospital, or Hunan Provincial Cancer Hospital. |
| Ethics oversight | This study was approved by the Ethics Committee of Chongqing University Three Gorges Hospital. |

Note that full information on the approval of the study protocol must also be provided in the manuscript.

# Field-specific reporting

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☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

# Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

| | |
|---|---|
| Sample size | Sample sizes were chosen based on the availability of samples collected across nine cancer types. No specific statistical methods were used to predetermine sample size. The number of donors depends on whether vascular cells are detected in them. By in silico sorting, samples who passed the quality control for single-cell libraries and had high-quality vascular cells were included. Detailed clinical characteristics are summarized in Supplementary Table 1. The same standards are applied to public datasets. In total, 372 donors (66 in-house and 306 from public databases) and 437 samples (97 in-house and 340 from public databases) were included. Our approach ensured a good representation for pan-cancer analysis. |
| Data exclusions | Data was not excluded from analysis. |

| | |
|---|---|
| Replication | In the mIHC experiment, 3-5 fields of view were selected for each sample, and the image results were evaluated by professional pathologists. In the zebrafish experiment, we repeated the experiment twice and selected representative pictures. We have also included the number of biologically independent samples in the legend for each figure. |
| Randomization | Randomization was not relevant for this study as there were not multiple groups requiring randomization. |
| Blinding | No blinding was performed in this study because there were no experimental groups. |

# Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

## Materials & experimental systems

| n/a | Involved in the study |
|---|---|
| ☐ | ☒ Antibodies |
| ☐ | ☒ Eukaryotic cell lines |
| ☒ | ☐ Palaeontology and archaeology |
| ☐ | ☒ Animals and other organisms |
| ☒ | ☐ Clinical data |
| ☒ | ☐ Dual use research of concern |
| ☒ | ☐ Plants |

## Methods

| n/a | Involved in the study |
|---|---|
| ☒ | ☐ ChIP-seq |
| ☒ | ☐ Flow cytometry |
| ☒ | ☐ MRI-based neuroimaging |

## Antibodies

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|---|---|
| Antibodies used | Primary antibodies: CD31 (Abcam, Cat# ab28364, 1:100), APLN (Abcam, Cat# ab230536, 1:200), CK17 (Abcam, Cat# ab53707, 1:100), SPP1 (R&D SYSTEMS, Cat# AF1433, 1:100), CD68 (Abcam, Cat# ab213363, 1:8000), MCAM (CST, Cat# 81701T, 1:200), LYVE1 (Abcam, Cat# ab219556, 1:1000), NFKB1 (Abcam, Cat# ab305263, 1:500), TSC22D3 (ThermoFisher, Cat# PA5-98816, 1:500), BASP1 (Abcam, Cat# ab312322, 1:2000), PDGFB (Abcam, Cat# ab107101, 1:1000), PDGFRB (Abcam, Cat# ab313777, 1:500), ICAM1 (Abcam, Cat# ab109361, 1:500), SPN (Abcam, Cat# ab235452, 1:2000), and aSMA (CST, Cat# 19245T, 1:1000).<br>Secondary antibodies: 1X Opal Anti-Ms + Rb HRP or HRP-conjugated Rabbit anti-Goat IgG (H+L) (ABclonal, AS029, 1:1000) |
| Validation | All antibodies were validated on human breast cancer tissue. Among them, CD31 (Abcam, Cat# ab28364), CK17 (Abcam, Cat# ab53707), CD68 (Abcam, Cat# ab213363), LYVE1 (Abcam, Cat# ab219556), MCAM (CST, Cat# 81701T), SPP1 (R&D SYSTEMS, Cat# AF1433), APLN (Abcam, Cat# ab230536), BASP1 (Abcam, Cat# ab312322), PDGFB (Abcam, Cat# ab107101), PDGFRB (Abcam, Cat# ab313777), ICAM1 (Abcam, Cat# ab109361), SPN (Abcam, Cat# ab235452), and aSMA (CST, Cat# 19245T) were expressed in the cell membrane and cytoplasm. The NFKB1 (Abcam, Cat# ab305263) and TSC22D3 (ThermoFisher, Cat# PA5-98816) were expressed in the cytoplasm and nucleus. APLN (Abcam, Cat# ab230536) was localized in endothelial cells. NFKB1 (Abcam, Cat# ab305263) and TSC22D3 (ThermoFisher, Cat# PA5-98816) were localized in lymphatic endothelial cells. SPP1 (R&D SYSTEMS, Cat# AF1433) was localized in pericytes and macrophages. aSMA (CST, Cat# 19245T) was localized in fibroblasts. |

## Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

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| Cell line source(s) | A549, MCF7, PLC/PRF/5, BEAS-2B, and MCF-10A were purchased from American Type Culture Collection (ATCC). HCCLM3 was purchased from China Center for Type Culture Collection (CCTCC). |
| Authentication | All cell lines were identified to the correct species. |
| Mycoplasma contamination | All cell lines were routinely tested and were negative for Mycoplasma. |
| Commonly misidentified lines (See ICLAC register) | No commonly misidentified lines were used in the study. |

## Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

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| Laboratory animals | Transgenic Tg(flk1:GFP) zebrafish (AB strain) are described in detail in Lin Wen et al. 2021. The adult zebrafish were used for mating. The embryos were obtained for injection at 24 hours post-fertilization. Zebrafish are housed on a 14-hour light/10-hour dark cycle, with a temperature controlled at 28.5±1°C and a pH range of 7.0-7.8. All zebrafish embryos were tested at 24 hours post-fertilization (hpf). |

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| Wild animals | No wild animals were involved in this study. |
| Reporting on sex | Adult male zebrafish were mated with adult female zebrafish to obtain embryos, which were used for embryo injection experiments without distinguishing the sex of the embryos. |
| Field-collected samples | No samples collected from the field were included in this study. |
| Ethics oversight | All animal studies were performed under the protocol approved by the Ethics Committee of Chongqing University Three Gorges Hospital. |

Note that full information on the approval of the study protocol must also be provided in the manuscript.